

UNIT-2

MT-2

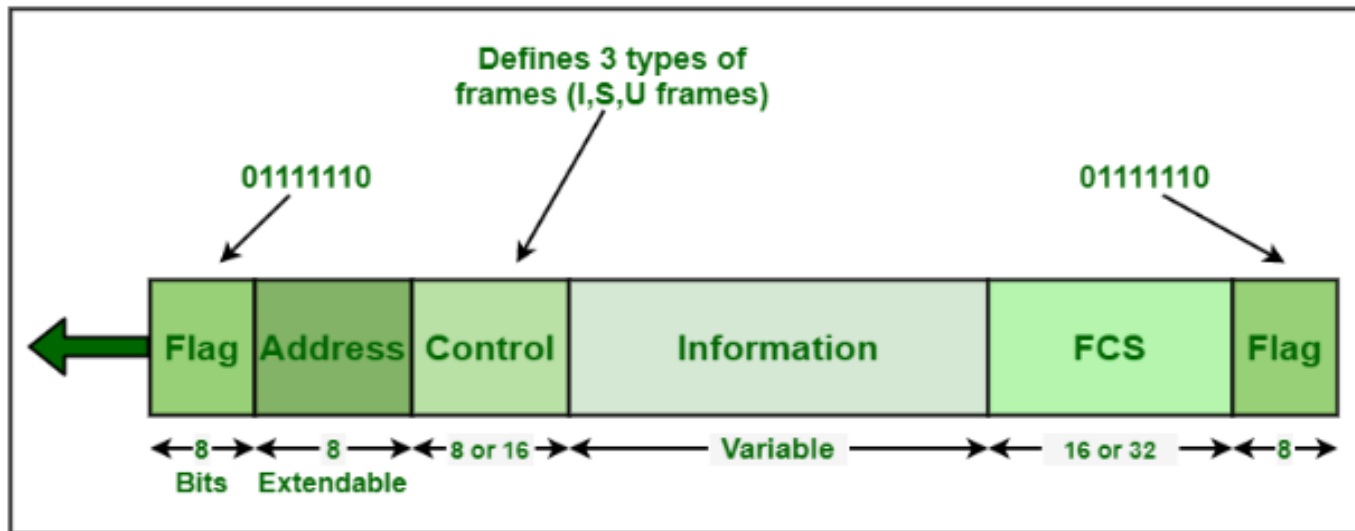
- **HDLC (High-Level Data Link Control)** is a **bit-oriented protocol** used for communication over **point-to-point** and **multipoint** links in computer networks.
- It's defined by **ISO (International Organization for Standardization)** and operates at the **Data Link Layer (Layer 2)** of the OSI model.

- **FeatureDescription**
- **Bit-oriented** Uses bit patterns for control information (unlike character-oriented).
- **Synchronous** Transmits data in a continuous stream, using clock synchronization.
- **Error Detection** Uses CRC (Cyclic Redundancy Check) to detect transmission errors.
- **Flow Control** Manages data flow between sender and receiver.
- **Framing** Defines clear frame boundaries using special flag sequences.

- When transferring data, stations are in one of three modes:
- **Normal Response Mode (NRM)** where the secondary station needs permission from the primary station before it can transmit data. Mainly used on multi-point lines.
- **Asynchronous Response Mode (ARM)** where the secondary station can send data without receiving permission from the primary station. This is hardly ever used.
- **Asynchronous Balanced Mode (ABM)** where either station can initiate transmission without permission from the other. This is the most common mode used on point-to-point links.

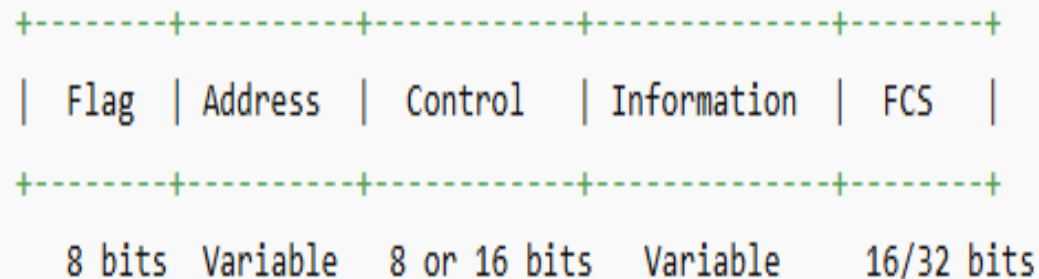
- Size of Different Fields :**

Field Name	Size (bits)
Flag	8 bits
Address	Field 8 bits
Control	Field 8 or 16 bits
Information	Field Variable (not used in some type of HDLC frames)
FCS (Frame Check Sequence)	Field 16 or 32 bits
Closing Flag	Field 8 bits



Basic Frame Structure

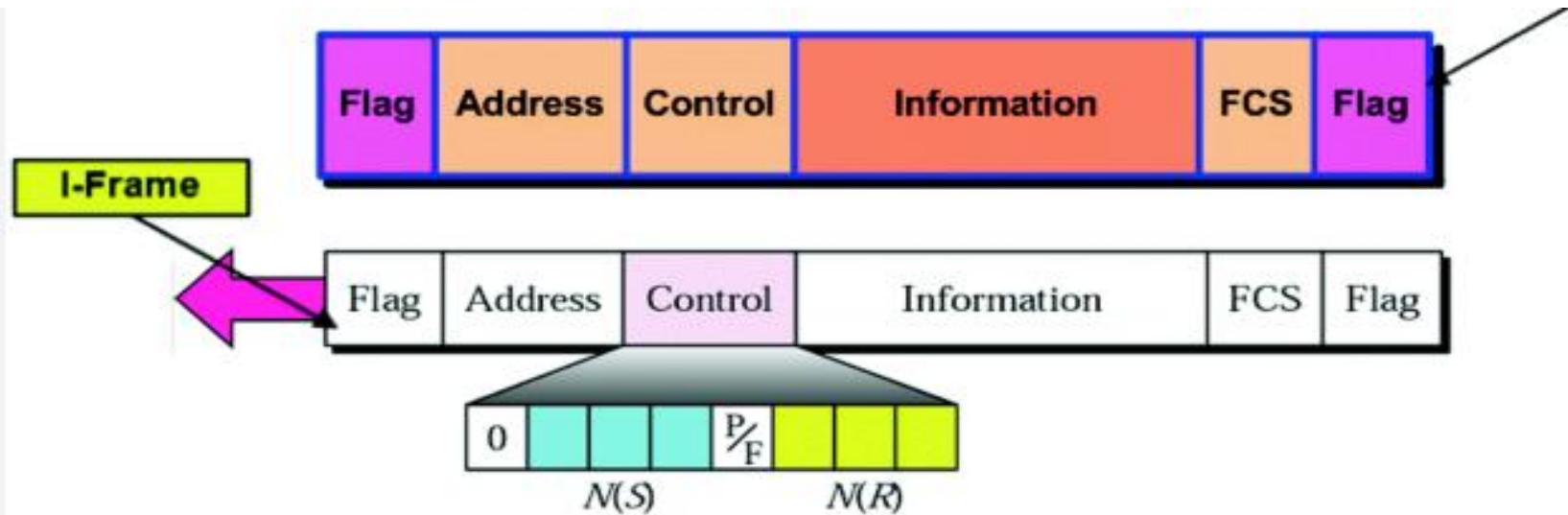
- HDLC Frame Structure



- **Flag:** Marks beginning and end of the frame (01111110)
- **Address:** Identifies the destination (used in multipoint configurations)
- **Control:** Manages frame type and flow control
- **Information:** Carries user data (only in information frames)
- **FCS (Frame Check Sequence):** Used for error detection

- **Types of HDLC Frames:**
- **I-Frames (Information Frames)**
 - Carry user data and control info (like sequence numbers)
 - Used in normal data transmission
- **S-Frames (Supervisory Frames)**
 - Used for flow and error control (e.g., ACKs, NAKs)
- **U-Frames (Unnumbered Frames)**
 - Used for link management (e.g., connection setup, disconnection)

- The following diagram details the HDLC frame format:



- **I-Frames (Information Frames)**

- **Purpose:**

Used for **transmitting user data** along with control information such as **sequence numbers** (for reliable transmission).

- **Contains:**

- Send Sequence Number (**N(S)**)
- Receive Sequence Number (**N(R)**)
- **Poll/Final (P/F)** bit (used for control)

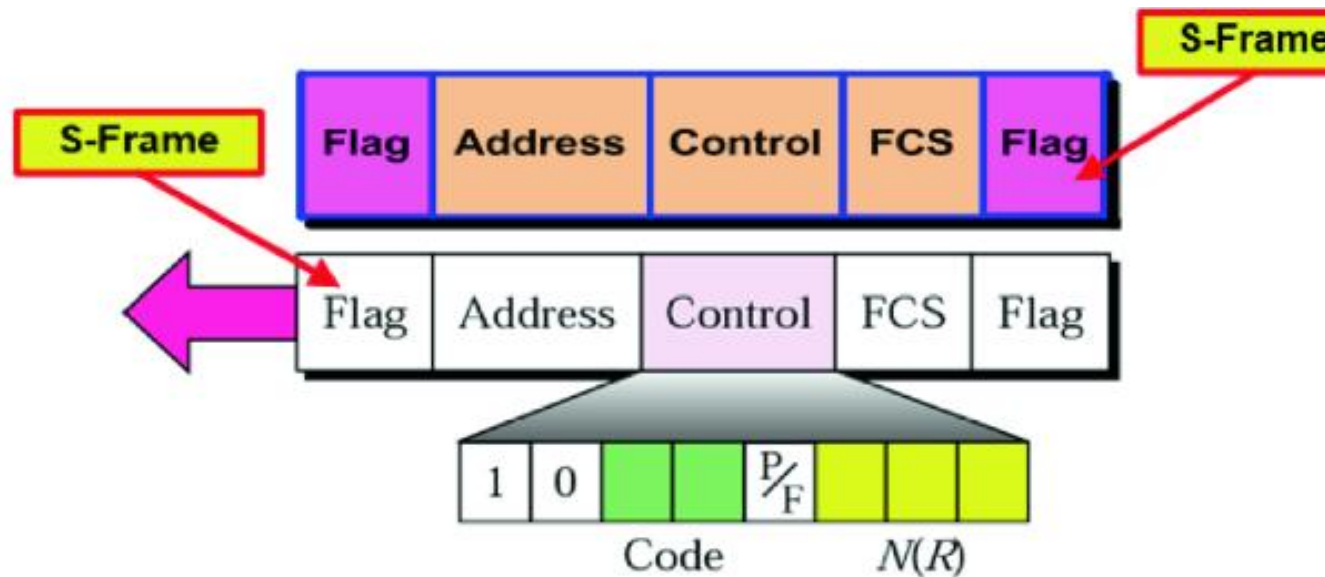
- **Used In:**

- **Data transfer**
- **Acknowledgment of received frames**

- **Example Use:**

Device A sends a file to Device B over HDLC; I-Frames carry the file's data and track sequence numbers.

- S Frame



00 – RR (Receive Ready)

01 – RNR (Receive Not Ready)

10 – REJ (Reject) – reject

11 – SREJ – selective reject

$N(R)$ – request sequence number

- **S-Frames (Supervisory Frames)**

- **Purpose:**

Used for **flow control and error control**—they **don't carry user data**, only **ACKs/NAKs** and other control signals.

- **Types:**

- **RR (Receive Ready):** Acknowledges receipt and ready for more
- **RNR (Receive Not Ready):** Acknowledges receipt but not ready for more
- **REJ (Reject):** Negative ACK, ask for retransmission from a certain frame
- **SREJ (Selective Reject):** Request retransmission of a specific frame

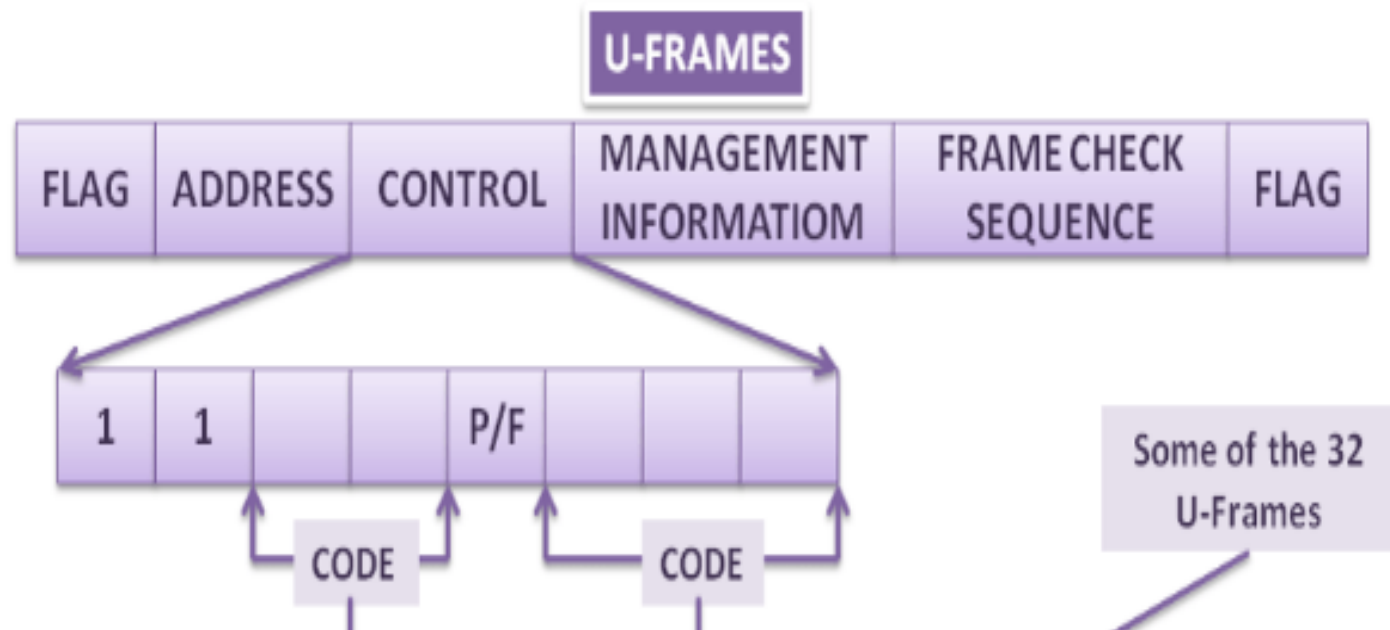
- **Contains:**

- **Receive Sequence Number (N(R))**
- **P/F bit**
- Frame type identifier (to distinguish RR, RNR, etc.)

- **Example Use:**

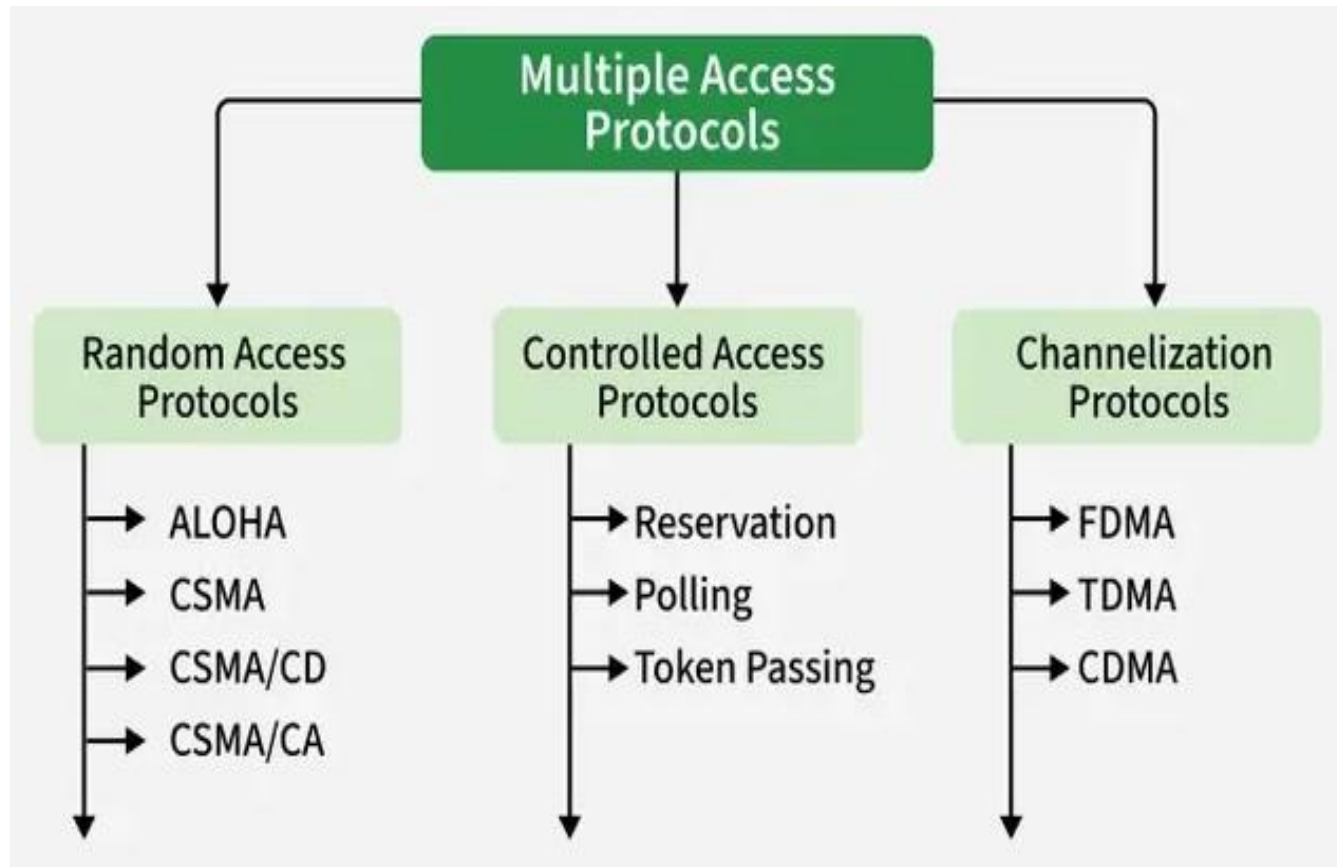
Device B receives data and sends an RR frame to let Device A know it can send more.

- Un Numbered



- **U-Frames (Unnumbered Frames)**
- **Purpose:**
Used for **control purposes** like **link setup, disconnection,** and **mode setting**. These frames are not numbered and are **not used for data transfer**.
- **Contains:**
- Control info only (no sequence numbers)
- Various commands like:
 - **SABM** (Set Asynchronous Balanced Mode)
 - **DISC** (Disconnect)
 - **UA** (Unnumbered Acknowledge)
 - **FRMR** (Frame Reject)
- **Example Use:**
Before sending data, Device A sends a **SABM** to start the communication. Device B replies with **UA** to confirm.

- **MEDIUM ACCESS CONTROL SUBLAYER**
- The MAC sublayer is the bottom part of the data link layer. The protocols used to determine who goes next on a multiaccess channel belong to a sublayer of the data link layer called the MAC (Medium Access Control) sublayer.
- The MAC sublayer is especially important in LANs, particularly wireless ones because wireless is naturally a broadcast channel. broadcast channels are sometimes referred to as multi-access channels or random access channels.



- **RANDOM ACCESS PROTOCOLS**
- In a random access protocol, a transmitting node always transmits at the full rate of the channel, namely, R bps.
- When there is a collision, each node involved in the collision repeatedly retransmits its frame(that is ,packet) until the frame gets through without a collision. But when a node experiences a collision, it doesn't necessarily retransmitting the frame right away. Instead it waits a random delay before retransmitting the frame.

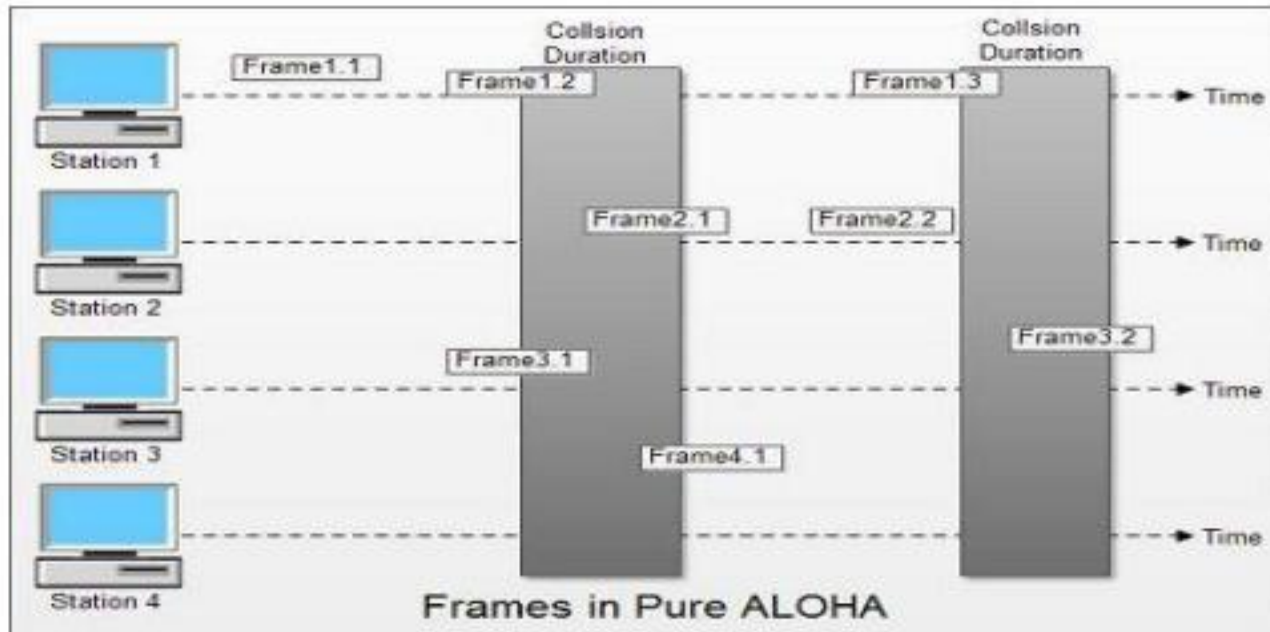
The most commonly used random access protocols

1. The ALOHA protocol
2. CSMA (carrier sense multiple access) protocol
3. CSMA/CD (carrier sense multiple access /collision detection) protocol and
4. Collision-Free Protocols
5. Limited-Contention Protocols

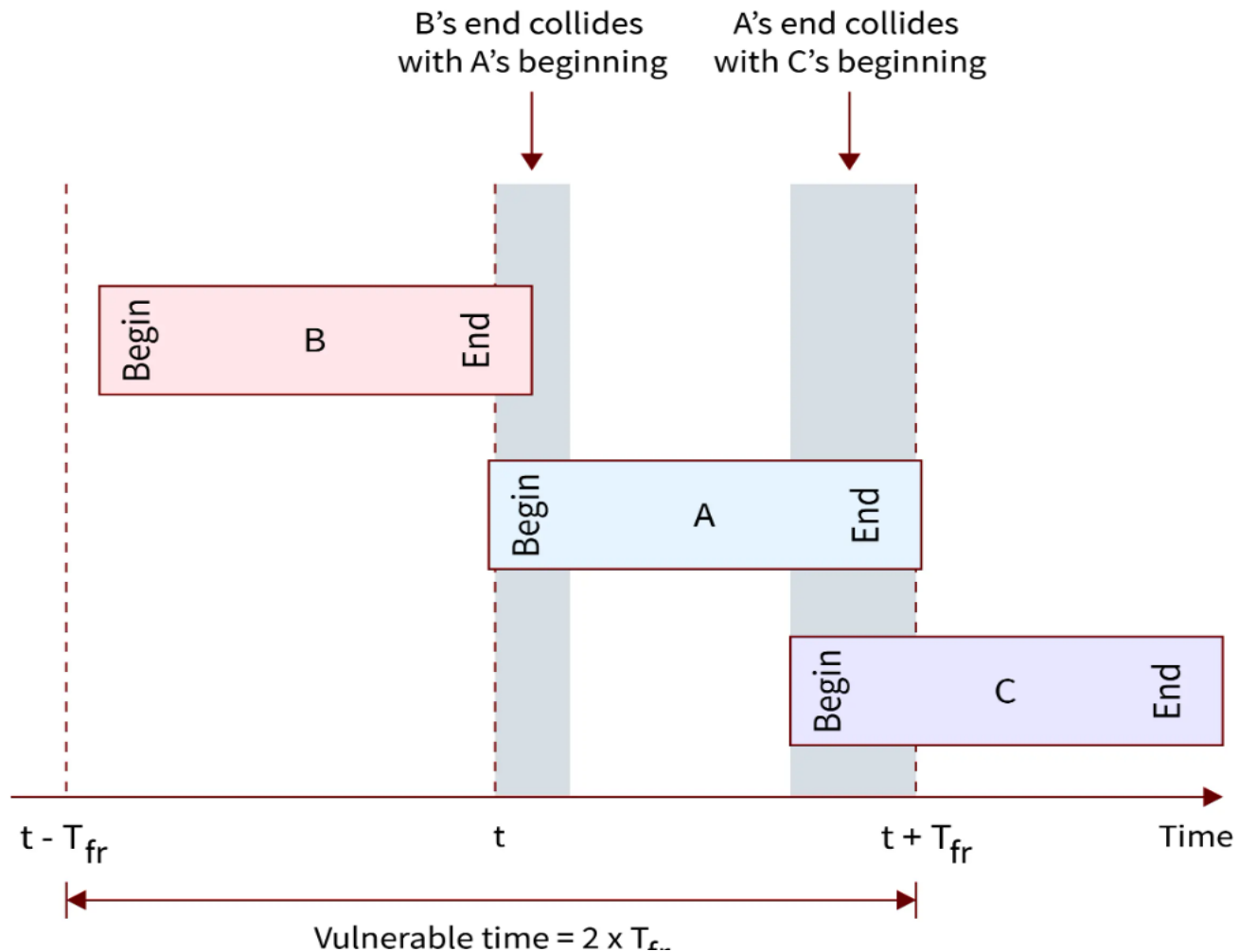
- **ALOHA**
- In the 1970s, Norman Abramson and his colleagues at the University of Hawaii devised a new and elegant method to solve the channel allocation problem.
- Although Abramson's work, called the ALOHA system, used ground-based radio broadcasting, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel.
- The two versions of ALOHA here: pure and slotted

- Pure ALOHA • In pure ALOHA, the stations transmit frames whenever they have data to send. • When two or more stations transmit simultaneously, there is collision and the frames are destroyed. • In pure ALOHA, whenever any station transmits a frame, it expects the acknowledgement from the receiver.

• Figure shows an example of frame collisions in pure ALOHA.

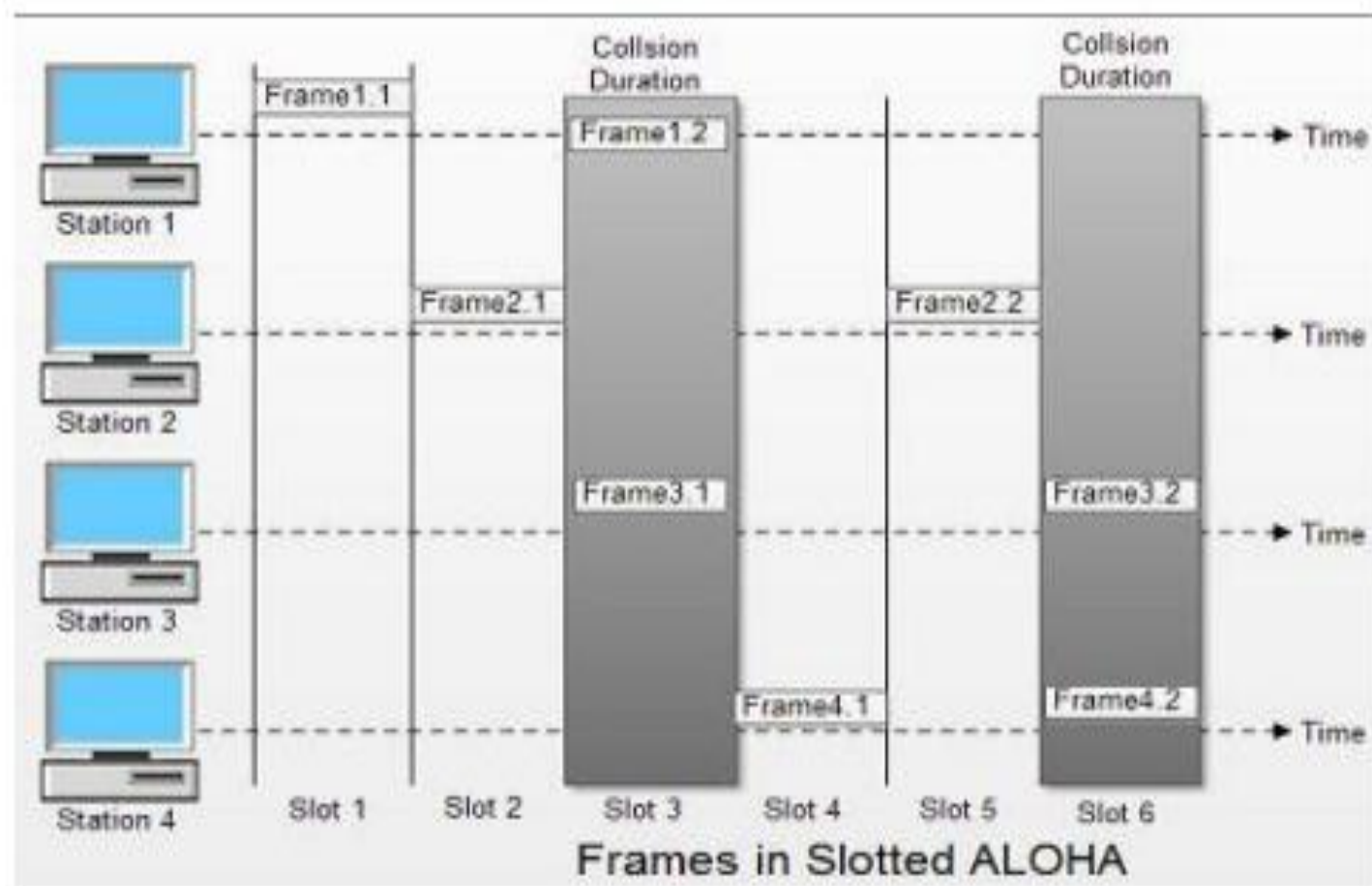


- **Vulnerable Time for Pure Aloha**

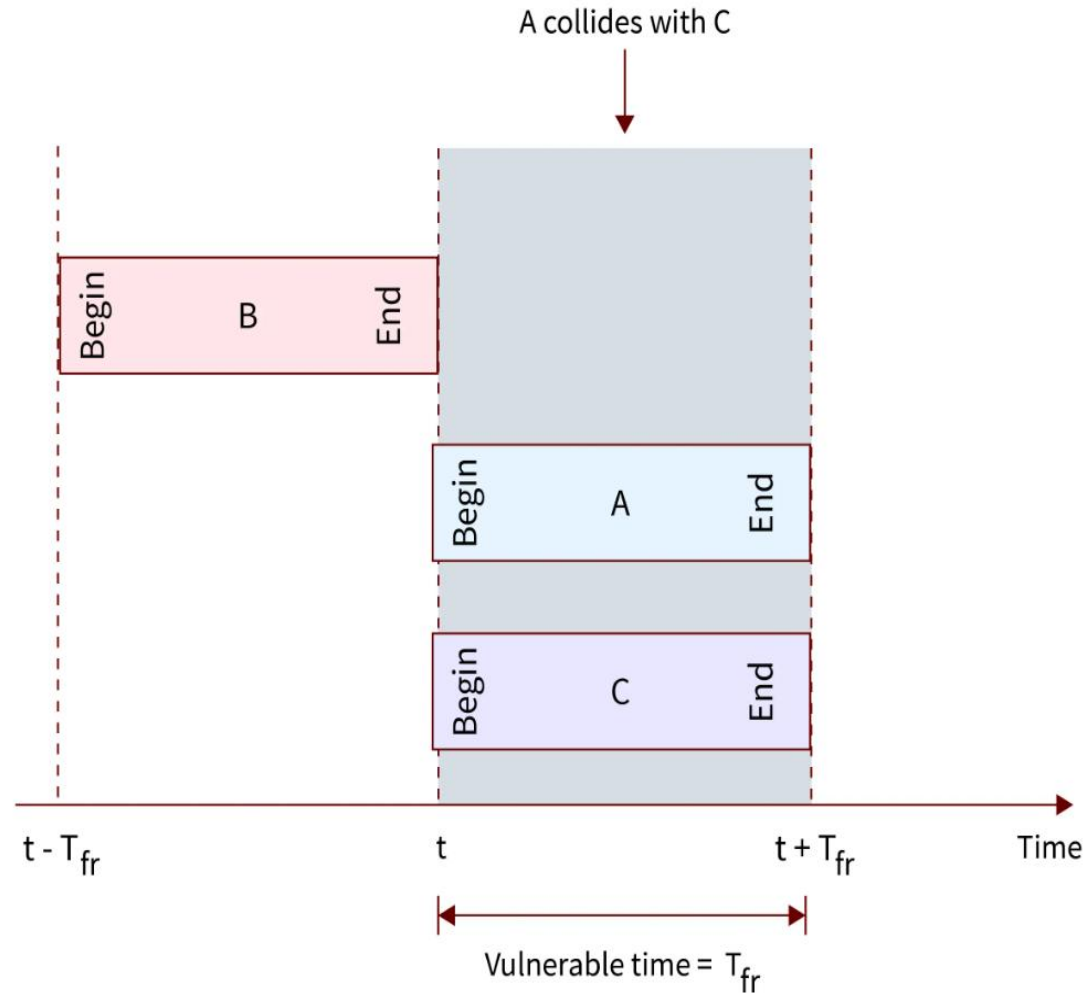


Slotted ALOHA

- Slotted ALOHA was invented to improve the efficiency of pure ALOHA as chances of collision in pure ALOHA are very high.
- In slotted ALOHA, the time of the shared channel is divided into discrete intervals called slots.
- The stations can send a frame only at the beginning of the slot and only one frame is sent in each slot.



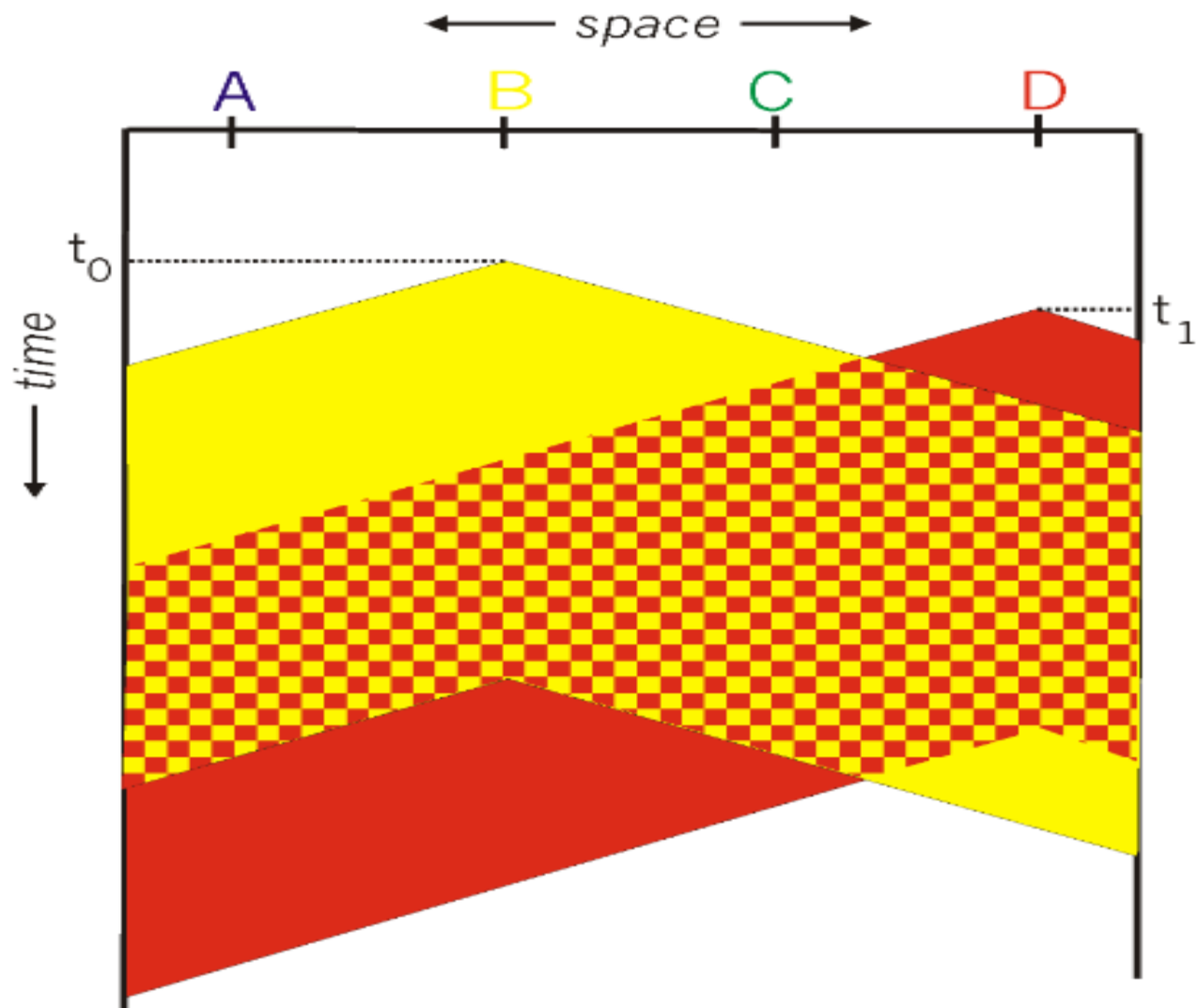
- Slotted ALOHA



- **CARRIER SENSE MULTIPLE ACCESS**

- Carrier Sense Multiple Access (CSMA) is a probabilistic Media Access Control (MAC) protocol in which a node verifies the absence of other traffic before transmitting on a shared transmission medium, such as an electrical bus, or a band of the electromagnetic spectrum.
- "Carrier Sense" describes the fact that a transmitter uses feedback from a receiver that detects a carrier wave before trying to send. That is, it tries to detect the presence of an encoded signal from another station before attempting to transmit.
- If a carrier is sensed, the station waits for the transmission in progress to finish before initiating its own transmission.

- CSMSA



- Carrier Sense Multiple Access (CSMA) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer. It senses or listens whether the shared channel for transmission is busy or not, and transmits if the channel is not busy.
- 1. **1-persistent CSMA** : In 1-persistent CSMA, the station continuously senses the channel to check its state i.e. idle or busy so that it can transfer data or not. In case when the channel is busy, the station will wait for the channel to become idle. When station found idle channel, it transmits the frame to the channel without any delay. It transmits the frame with probability 1. Due to probability 1, it is called 1-persistent CSMA.

- **1. 1-persistent CSMA**

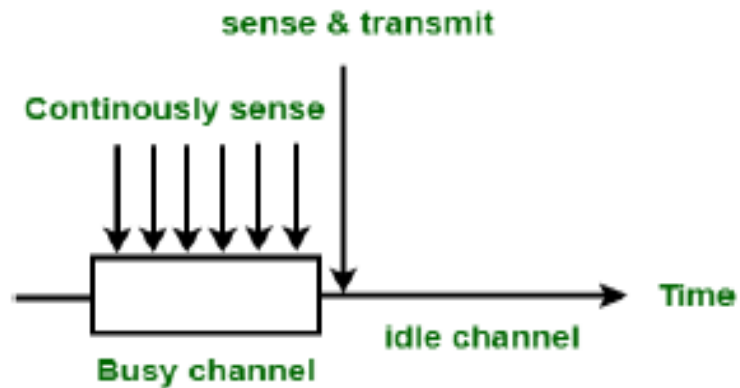
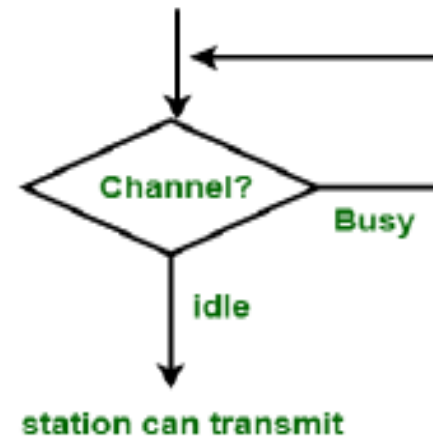


Figure – 1-persistent CSMA



- **Non-persistent CSMA** : In this method, the station that has frames to send, only that station senses for the channel. In case of an idle channel, it will send frame immediately to that channel.
- In case when the channel is found busy, it will wait for the random time and again sense for the state of the station whether idle or busy. In this method, the station does not immediately sense for the channel for only the purpose of capturing it when it detects the end of the previous transmission.
- The main advantage of using this method is that it reduces the chances of collision. The problem with this is that it reduces the efficiency of the network.

- **Non-persistent CSMA**

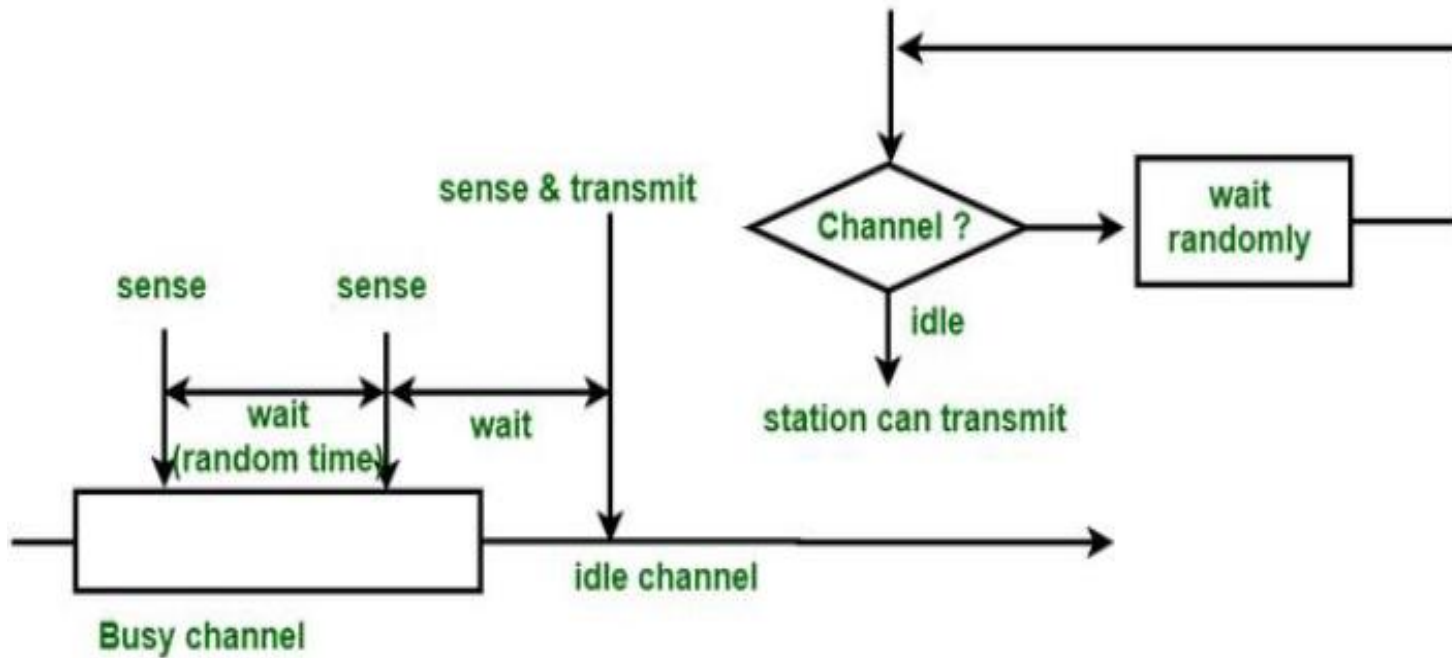


Figure – Non-persistent CSMA

- **3. p-persistent CSMA** : This is the method that is used when channel has time-slots and that time-slot duration is equal to or greater than the maximum propagation delay time.
- When the station is ready to send the frames, it will sense the channel. If the channel found to be busy, the channel will wait for the next slot. If the channel found to be idle, it transmits the frame with probability p , thus for the left probability i.e. q which is equal to $1-p$ the station will wait for the beginning of the next time slot.
- In case, when the next slot is also found idle it will transmit or wait again with the probabilities p and q . This process is repeated until either the frame gets transmitted or another station has started transmitting.

- p-persistent CSMA

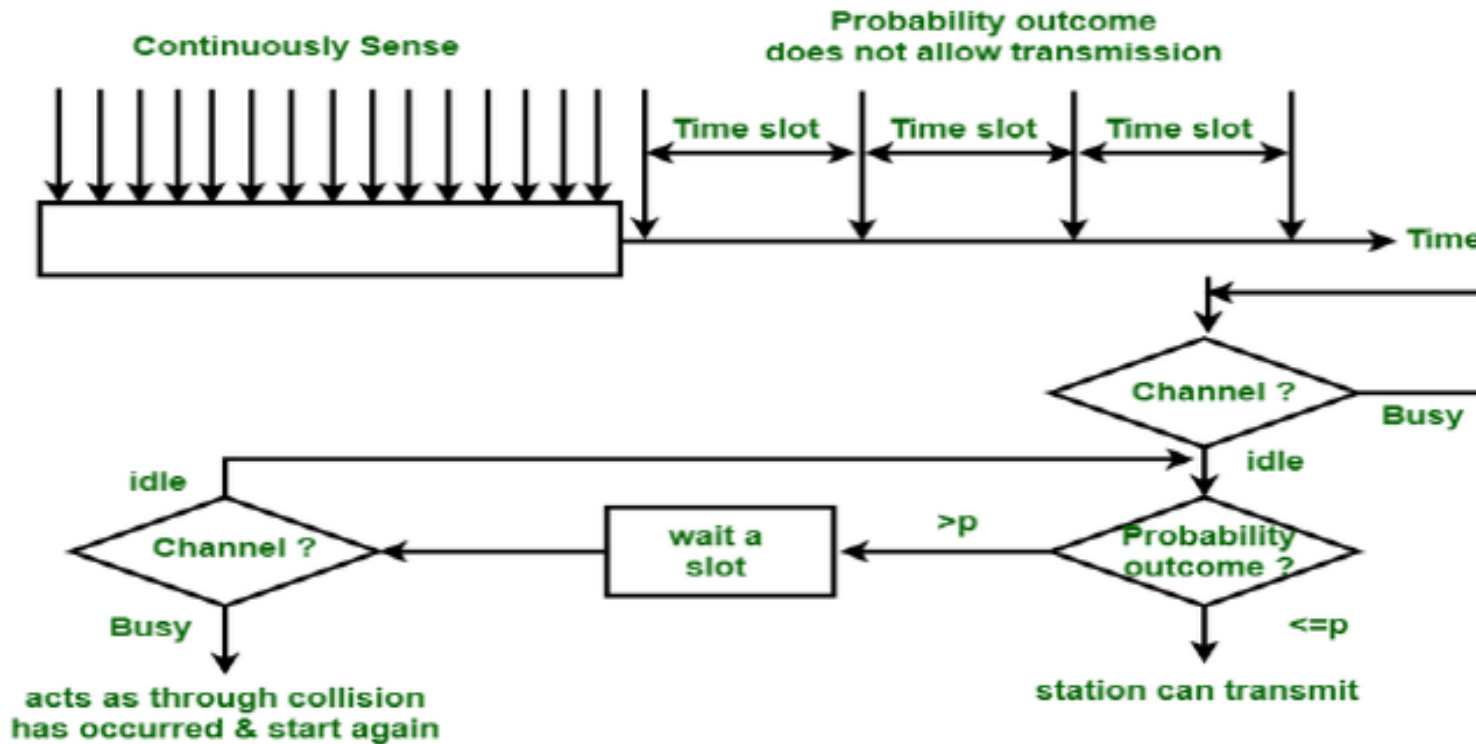
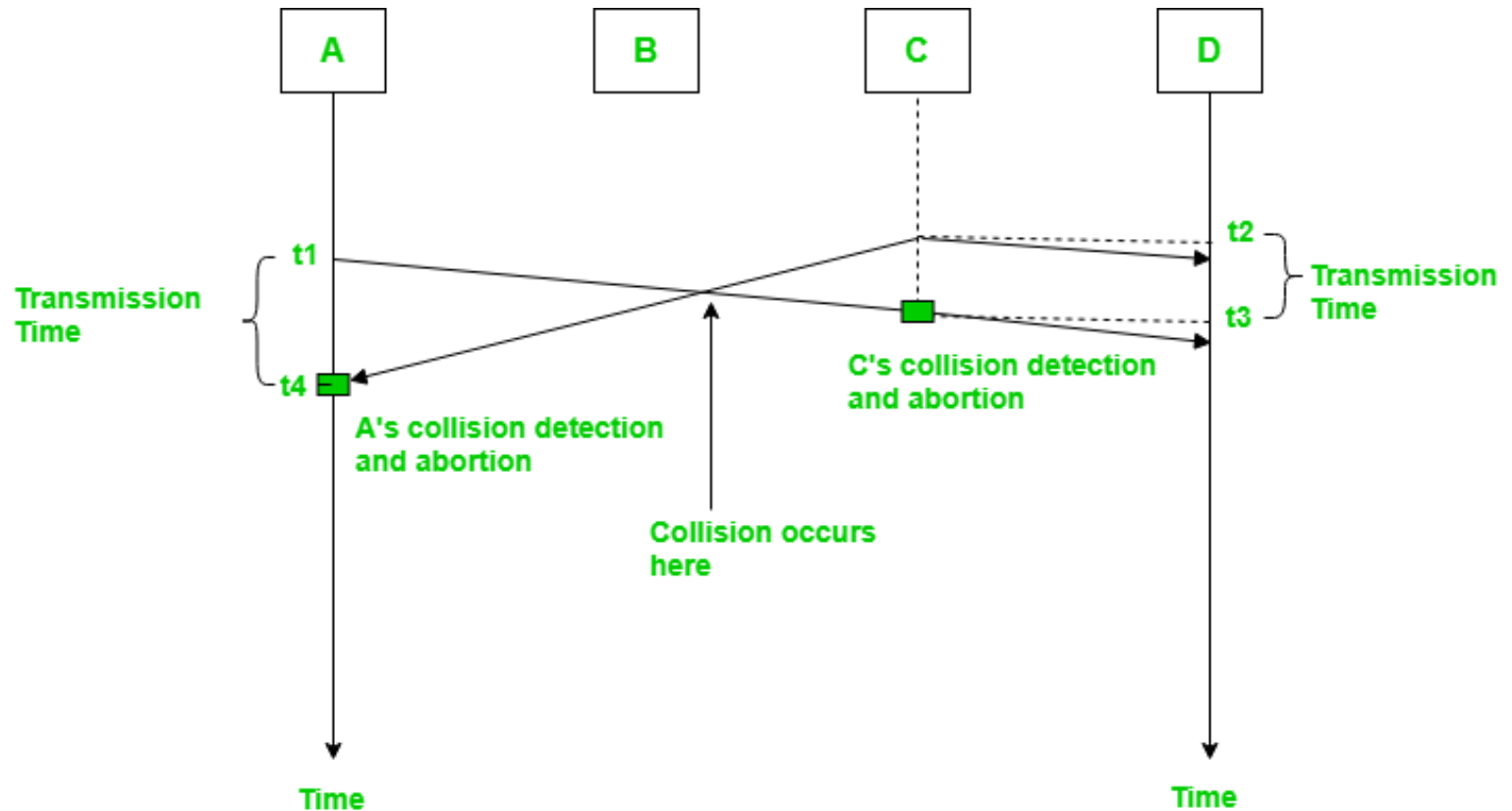


Figure – p-persistent CSMA

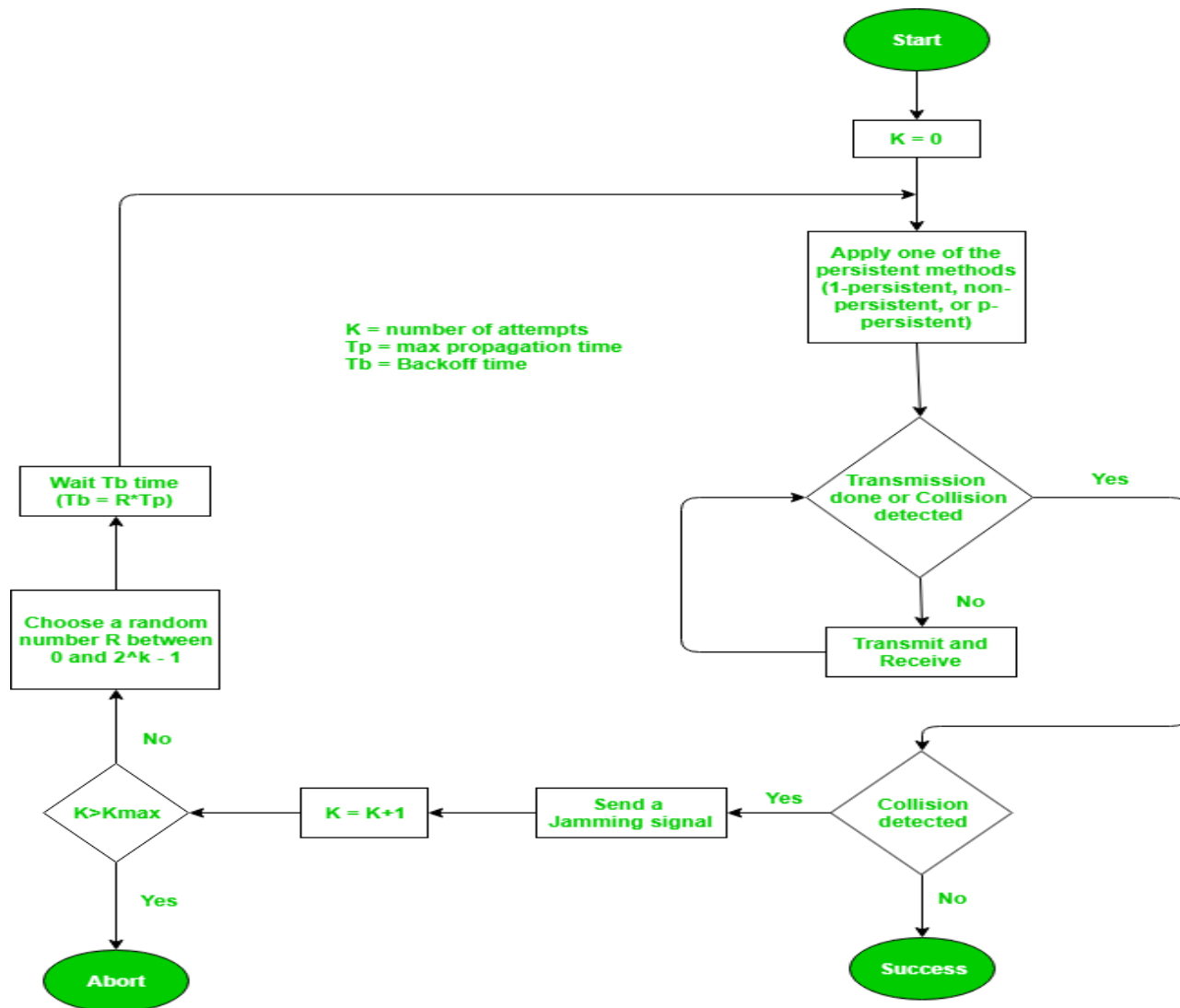
- **CSMA/CD**
- This method was developed to decrease the chances of collisions when two or more stations start sending their signals over the datalink layer. Carrier Sense multiple access requires that each station **first check the state of the medium before sending.**
- **1. Carrier Sense Multiple Access with Collision Detection (CSMA/CD)**
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful.

- In the diagram, A starts send the first bit of its frame at t_1 and since C sees the channel idle at t_2 , starts sending its frame at t_2 . C detects A's frame at t_3 and aborts transmission. A detects C's frame at t_4 and aborts its transmission.
- Transmission time for C's frame is therefore $t_3 - t_2$ and for A's frame is $t_4 - t_1$.
- So, the frame transmission time (T_{fr}) should be at least twice the maximum propagation time (T_p). This can be deduced when the two stations involved in collision are maximum distance apart.
- Process – The entire process of collision detection can be explained as follows:

- If successful, the station is finished, if not, the frame is sent again.

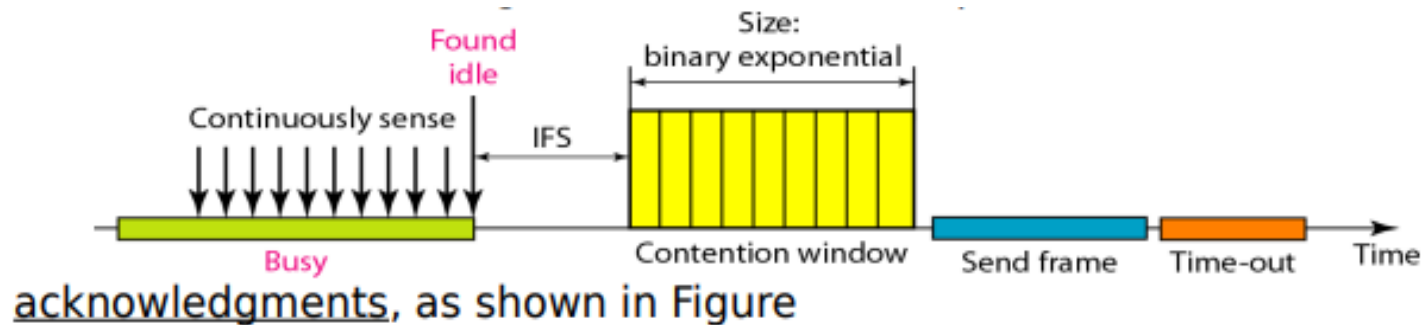


- CSMA/CD



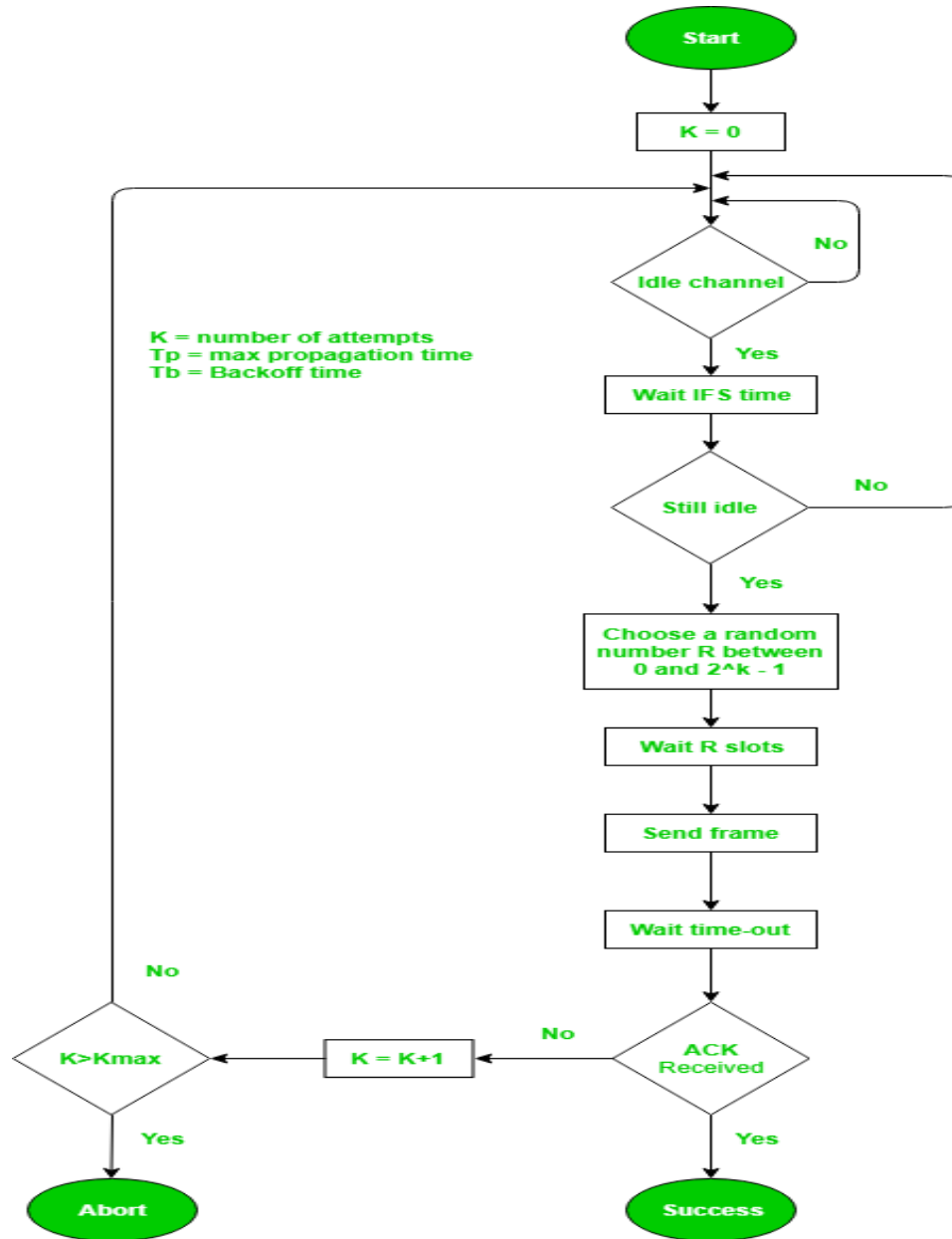
- **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) –**
- The basic idea behind CSMA/CA is that the station should be able to receive while transmitting to detect a collision from different stations.
- In wired networks, if a collision has occurred then the energy of received signal almost doubles and the station can sense the possibility of collision.
- In case of wireless networks, most of the energy is used for transmission and the energy of received signal increases by only 5-10% if a collision occurs. It can't be used by the station to sense collision. Therefore **CSMA/CA has been specially designed for wireless networks.**

- **These are three types of strategies:**
- 1. InterFrame Space (IFS) – When a station finds the channel busy, it waits for a period of time called IFS time. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority.
- 2. Contention Window – It is the amount of time divided into slots. A station which is ready to send frames chooses random number of slots as wait time.
- 3. Acknowledgements – The positive acknowledgements and time-out timer can help guarantee a successful transmission of the frame.



Timing in CSMA/CA

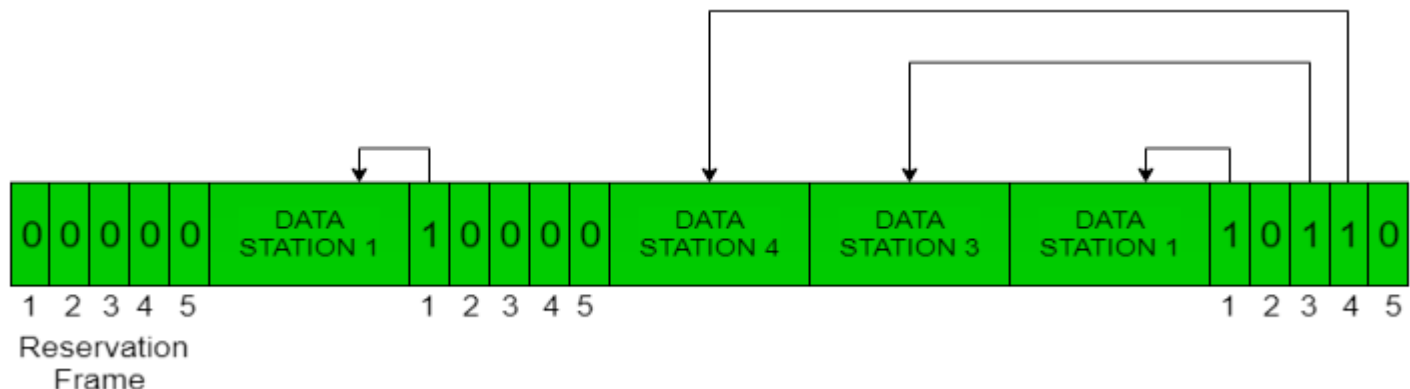
- CSMA/CA



- Controlled Access Protocols In controlled access
- The stations seek information from one another to find which station has the right to send.
- It allows only one node to send at a time, to avoid collision of messages on shared medium. The three controlled-access methods are:
 - 1 Reservation
 - 2 Polling
 - 3 Token Passing

- Reservation

- In the reservation method, a station needs to make a reservation before sending data. The time line has two kinds of periods: 1. Reservation interval of fixed time length 2. Data transmission period of variable frames.
- If there are M stations, the reservation interval is divided into M slots, and each station has one slot. Suppose if station 1 has a frame to send, it transmits 1 bit during the slot 1. No other station is allowed to transmit during this slot.

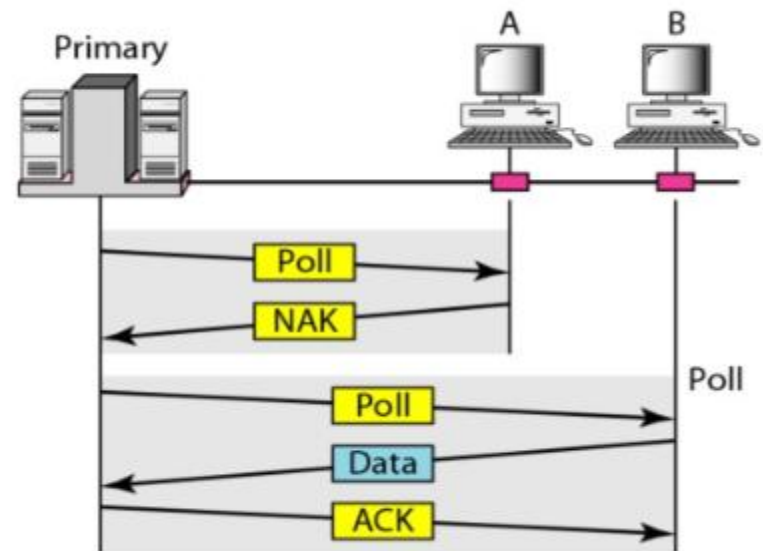
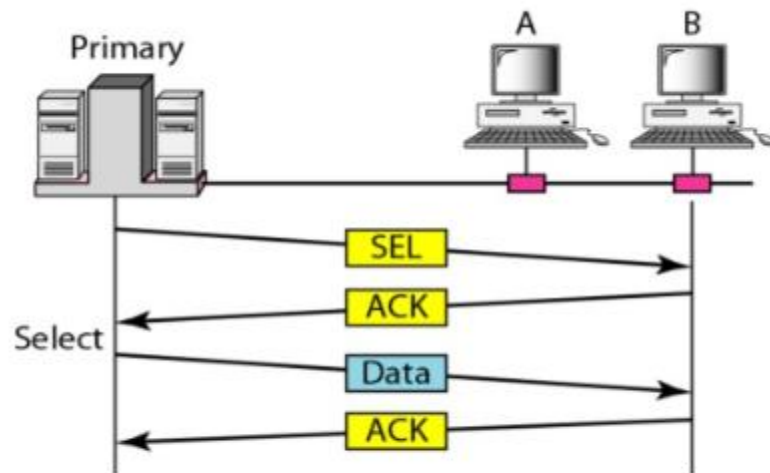


- In general, i th station may announce that it has a frame to send by inserting a 1 bit into i th slot. After all N slots have been checked, each station knows which stations wish to transmit.
- The stations which have reserved their slots transfer their frames in that order. After data transmission period, next reservation interval begins.
- Since everyone agrees on who goes next, there will never be any collisions. The following figure shows a situation with five stations and a five slot reservation frame.
- In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.

- Polling

- Polling process is similar to the roll-call performed in class. Just like the teacher, a controller sends a message to each node in turn.
- In this, one acts as a primary station(controller) and the others are secondary stations. All data exchanges must be made through the controller.
- The message sent by the controller contains the address of the node being selected for granting access. Although all nodes receive the message but the addressed one responds to it and sends data, if any.
- If there is no data, usually a “poll reject”(NAK) message is sent back. Problems include high overhead of the polling messages and high dependence on the reliability of the controller.

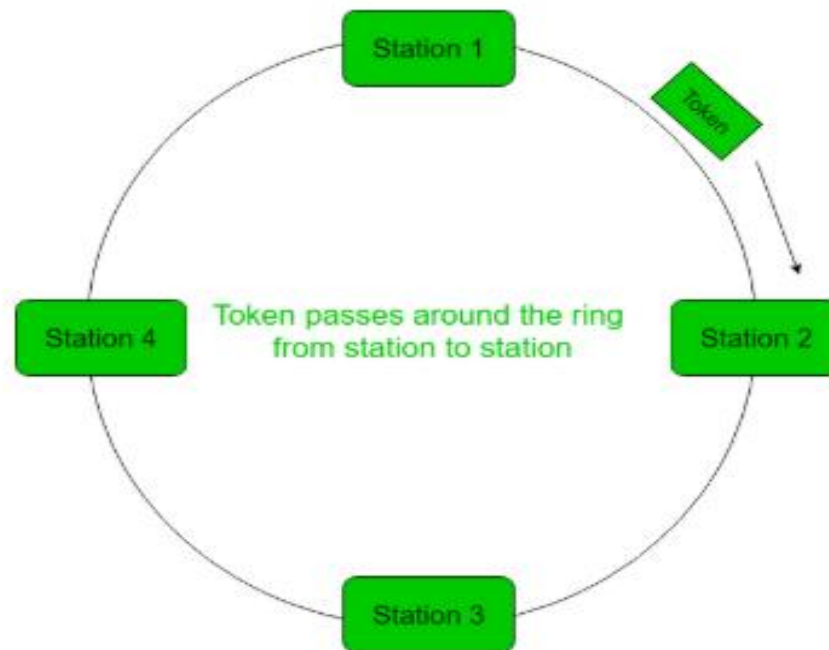
- Polling



- **Token Passing**

- In token passing scheme, the stations are connected logically to each other in form of ring and access of stations is governed by tokens. A token is a special bit pattern or a small message, which circulate from one station to the next in the some predefined order.
- In Token ring, token is passed from one station to another adjacent station in the ring whereas incase of Token bus, each station uses the bus to send the token to the next station in some predefined order.
- In both cases, token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station. If it has no queued frame, it passes the token simply.

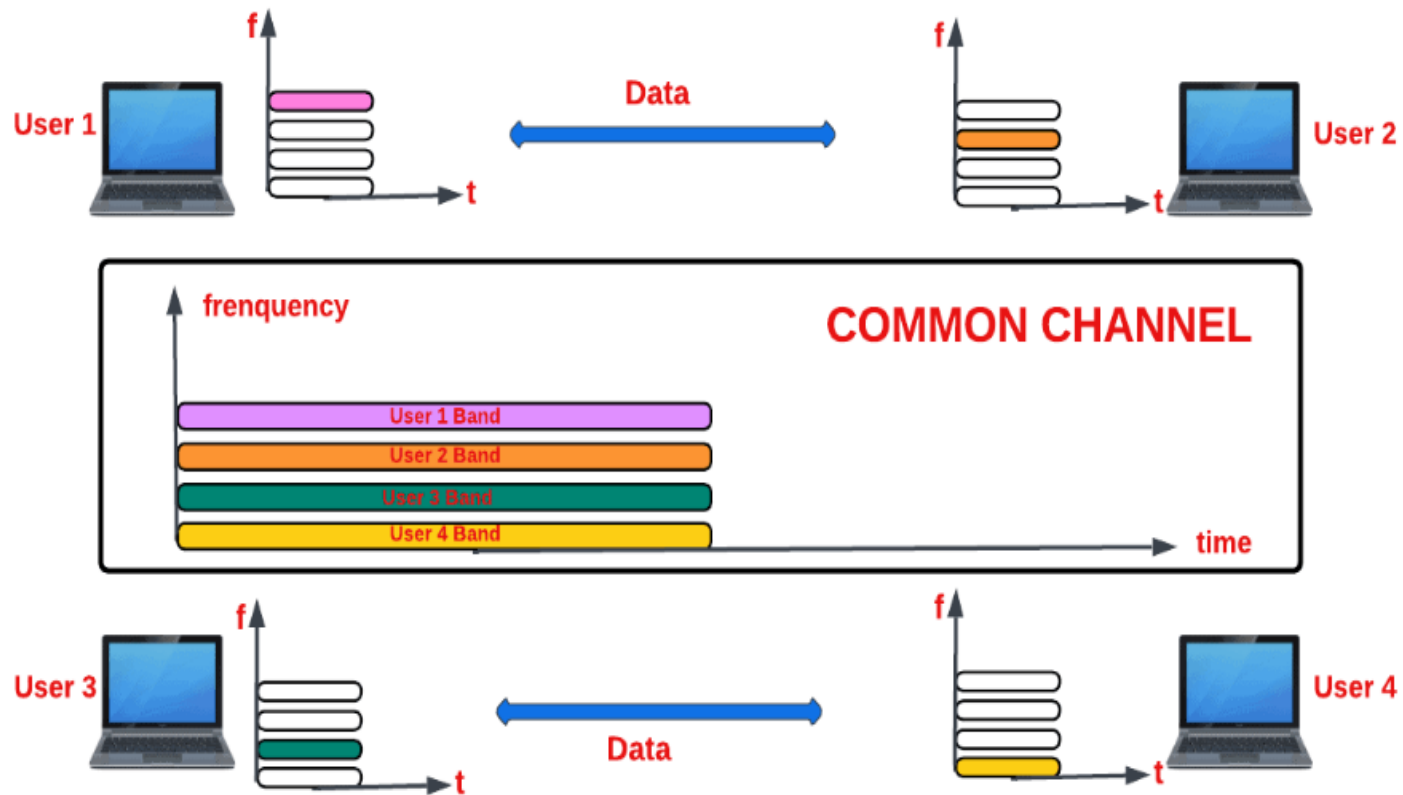
- After sending a frame, each station must wait for all N stations (including itself) to send the token to their neighbors and the other $N - 1$ stations to send a frame, if they have one. There exists problems like duplication of token or token is lost or insertion of new station, removal of a station, which need be tackled for correct and reliable operation of this scheme.



- Channels are communication resources we assign to a user to establish communication with other users in the network. These resources include frequency, time, code, space, [bandwidth](#), and so on. We name each channelization protocol based on the kind of channel resources we employ for channelization. These protocols are:
 - Frequency division multiple access (FDMA)
 - Time division multiple access (TDMA)
 - Code division multiple access (CDMA)

- **Frequency Division Multiple Access (FDMA)**
- **Protocol** We design the FDMA protocol by utilizing the channel resource of frequency. **By dividing the channel's bandwidth into non-overlapping frequency bands or slices, we allocate each user a distinct band for exclusive transmission and reception.** This enables the concurrent operation of multiple users within a single communication channel. To mitigate interference and crosstalk during data transmission, we equip each user with a customized bandpass filter complemented by a small guard band.
- The following figure depicts the FDMA protocol scenario:

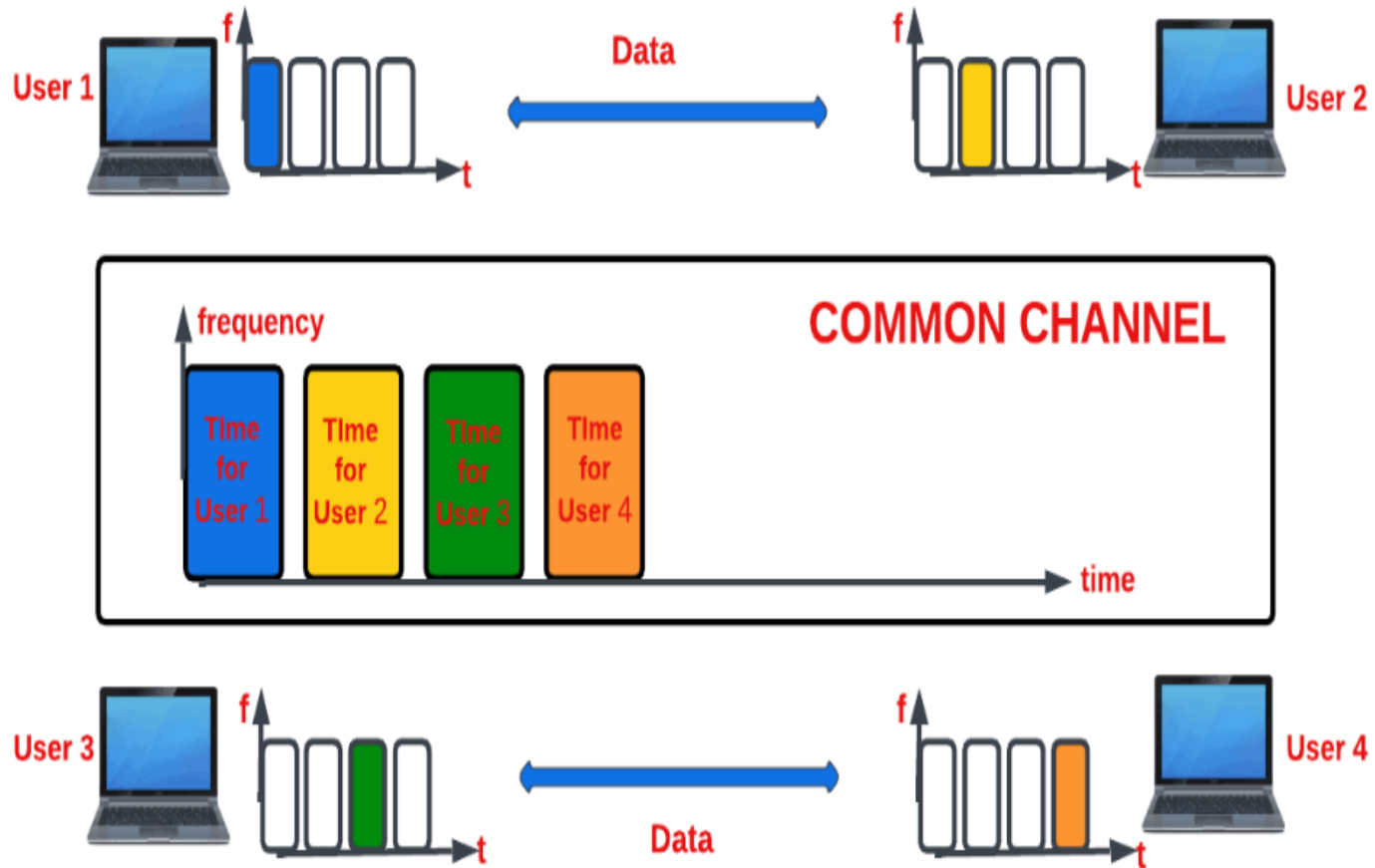
The following figure depicts the FDMA protocol scenario:



As seen in the above diagram, four users are attempting to access the same channel. To accommodate the four users without interfering with one another, we utilize FDMA to split the channel into four frequency bands.

- **Time Division Multiple Access (TDMA) Protocol**
- Time serves as the channel resource for this protocol. Using the TDMA protocol, we divide each user's signal into a different time slot and assign each time slot to a different user.
- This technique enables numerous users to share and use the same frequency channel. The global system of mobile communication (GSM) system, a 2G cellular system, is one significant application where we have used the TDMA protocol.
- Synchronizing the various users of the channel is one of the biggest challenges we face when utilizing the TDMA protocol. To achieve synchronization, we ensure each user knows their respective slots' start and endpoints. This is difficult to achieve, especially if the users are spread over a large area. To minimize this challenge, we insert guard times to each user channel.

A typical TDMA protocol looks like this:



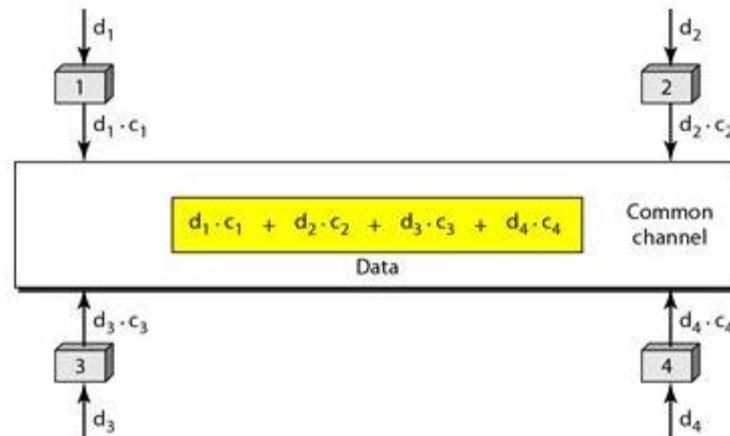
- **Code Division Multiple Access (CDMA) Protocol**

- The channel resource for this protocol is code. In the CDMA protocol, we allow multiple users to simultaneously transmit their data signals over the entire bandwidth of the common channel by assigning unique spreading codes to each user. This technique helps us utilize the channel more effectively.
- Let's see what a CDMA protocol looks like: The beauty of this protocol is that we don't share the frequency as we do in FDMA, and neither do we share time as we do in TDMA.
- In CDMA, we employ both time and frequency for users to send their data over the entire bandwidth of the channel with a unique spreading code allotted to the user. Only receivers having this unique code can receive the sent messages.

- **Implementation:**

Let us assume we have four stations 1, 2, 3, and 4 connected to the same channel. The data from station 1 are d_1 , from station 2 are d_2 , and so on. The code assigned to the first station is c_1 , to the second is c_2 , and so on. We assume that the assigned codes have two properties.

- 1. If we multiply each code by another, we get 0.
- 2. If we multiply each code by itself, we get 4 (the number of stations).
- With these two properties in mind, how the above four stations can send data using the same common channel, as shown in the following figure.



- Station 1 multiplies (a special kind of multiplication, as we will see) its data by its code to get $d1.c1$. Station 2 multiplies its data by its code to get $d2.c2$. And so on. The data that go on the channel are the sum of all these terms, as shown in the box.
-
- Any station that wants to receive data from one of the other three multiplies the data on the channel by the code of the sender. For example, suppose stations 1 and 2 are talking to each other. Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by $c1$ the code of station1. Because $(c1.c1)$ is 4, but $(c2 . c1)$, $(c3. c1)$, and $(c4 .c1)$ are all 0s, station 2 divides the result by 4 to get the data from station1.
- $data = (d1.c1 + d2.c2 + d3.c3 + d4.c4).c1$
- $= c1. d1. c1 + c1. d2. c2 + c1. d3. c3 + c1. d4. c4 = 4d1$

- **Code Division Multiple Access (CDMA) Protocol**

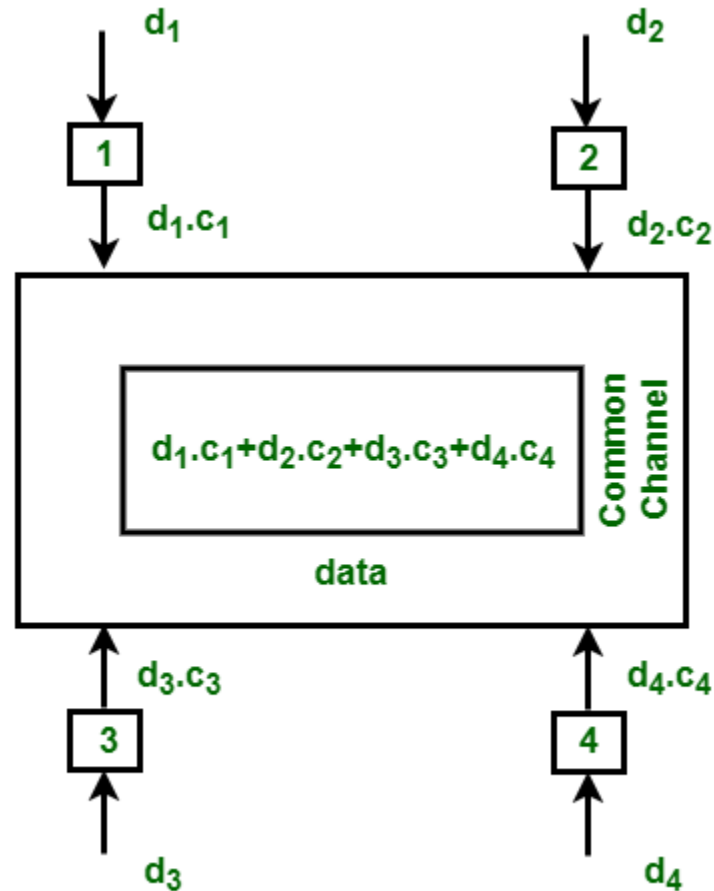


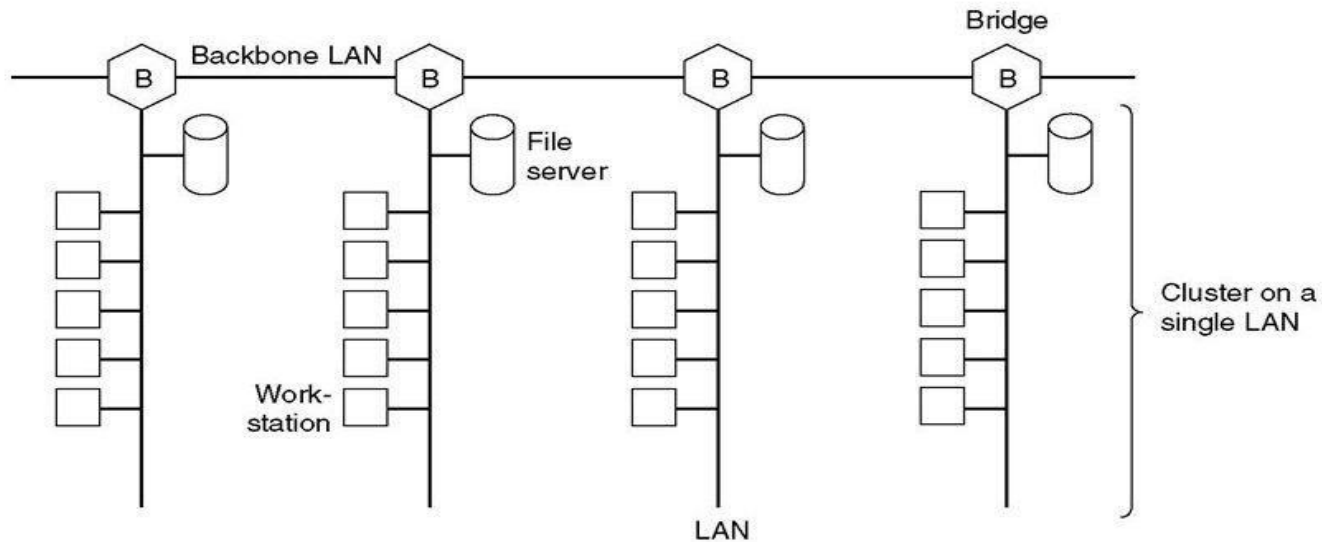
Figure - CDMA

- Assume 4 stations S1, S2, S3, S4. We'll use 4x4 Walsh Table to assign codes to them. $C1 = [+1 \ +1 \ +1 \ +1]$ $C2 = [+1 \ -1 \ +1 \ -1]$ $C3 = [+1 \ +1 \ -1 \ -1]$ $C4 = [+1 \ -1 \ -1 \ +1]$
- **Let their data bits currently be:**
 - $D1 = -1$
 - $D2 = -1$
 - $D3 = 0$ (Silent)
 - $D4 = +1$
- **Resultant channel sequence =**
 - $C1.D1 + C2.D2 + C3.D3 + C4.D4 = [-1 \ -1 \ -1 \ -1] + [-1 \ +1 \ -1 \ +1] + [0 \ 0 \ 0 \ 0] + [+1 \ -1 \ -1 \ +1] = [-1 \ -1 \ -3 \ +1]$
 - Now suppose station 1 wants to listen to station
 - Inner Product = $[-1 \ -1 \ -3 \ +1] \times C2 = -1 + 1 - 3 - 1 = -4$ Data bit that was sent = $-4/4 = -1$.

- **Data Link Layer Switching**

- Many organizations have multiple LANs and wish to connect them. LANs can be connected by devices called bridges, which operate in the data link layer. Bridges examine the data layer link addresses to do routing.
- Network switching is the process of forwarding data frames or packets from one port to another leading to data transmission from source to destination.
- Data link layer is the second layer of the Open System Interconnections (OSI) model whose function is to divide the stream of bits from physical layer into data frames and transmit the frames according to switching requirements.

Data Link Layer Switching

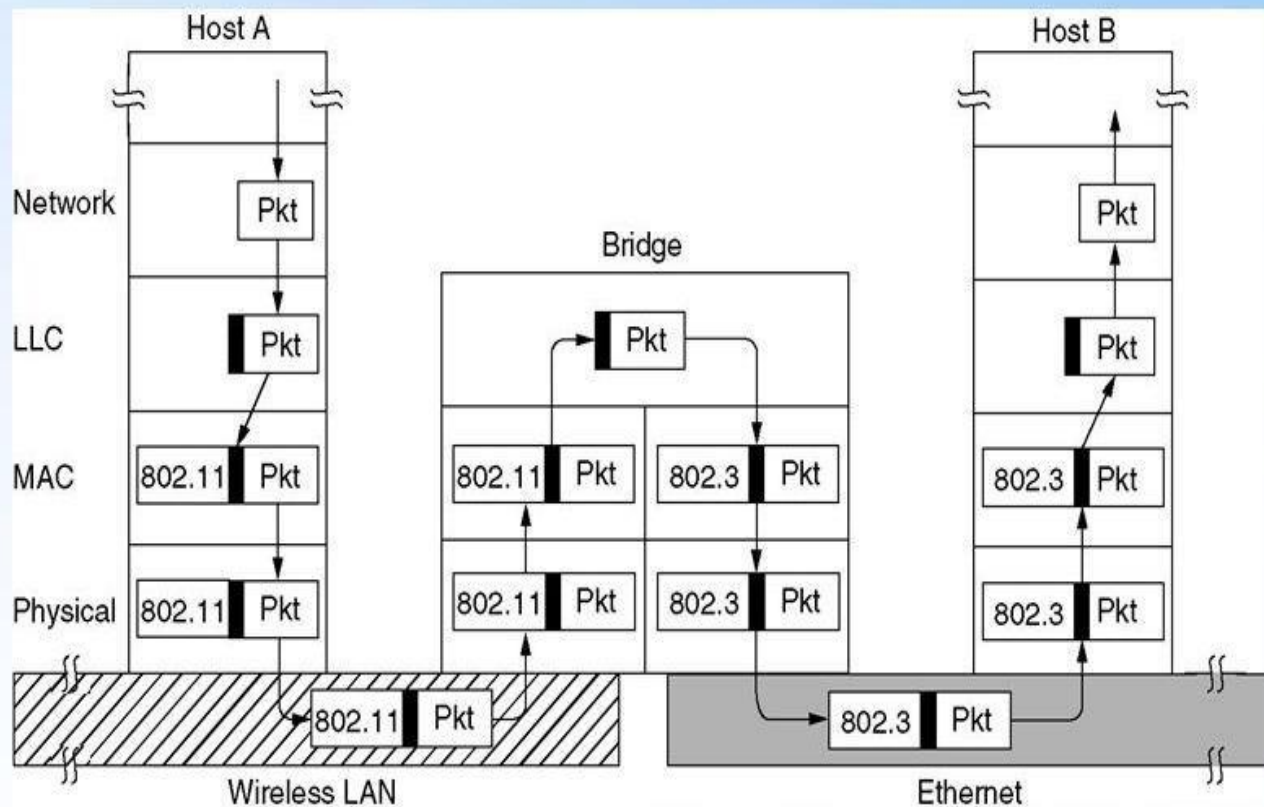


Multiple LANs connected by a backbone to handle a total load higher than the capacity of a single LAN.

- **Bridges from 802.x to 802.y**
- Having seen why bridges are needed, let us now turn to the question of how they work. Figure illustrates the operation of a simple two-port bridge.
- Host A on a wireless (802.11) LAN has a packet to send to a fixed host, B, on an (802.3) Ethernet to which the wireless LAN is connected.
- The packet descends into the LLC sublayer and acquires an LLC header (shown in black in the figure).
- Then it passes into the MAC sublayer and an 802.11 header is prepended to it (also a trailer, not shown in the figure). This unit goes out over the air and is picked up by the base station, which sees that it needs to go to the fixed Ethernet.
- When it hits the bridge connecting the 802.11 network to the 802.3 network, it starts in the physical layer and works.

From 802.x to 802.y

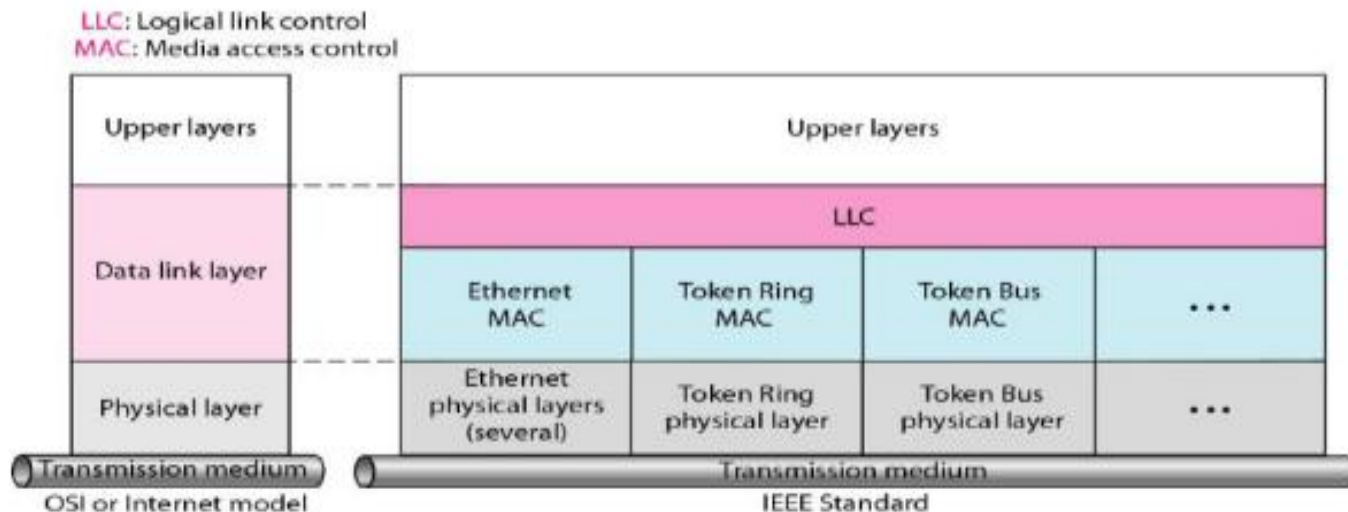
□ How do bridges work?



- In the MAC sublayer in the bridge, the 802.11 header is stripped off. The bare packet (with LLC header) is then handed off to the LLC sublayer in the bridge.
- In this example, the packet is destined for an 802.3 LAN, so it works its way down the 802.3 side of the bridge and off it goes on the Ethernet.
- Note that a bridge connecting k different LANs will have k different MAC sublayers and k different physical layers, one for each type.
- So far it looks like moving a frame from one LAN to another is easy. Such is not the case.
- In this section we will point out some of the difficulties that one encounters when trying to build a bridge between the various 802 LANs (and MANs). We will focus on 802.3, 802.11, and 802.16, but there are others as well, each with its unique problems.

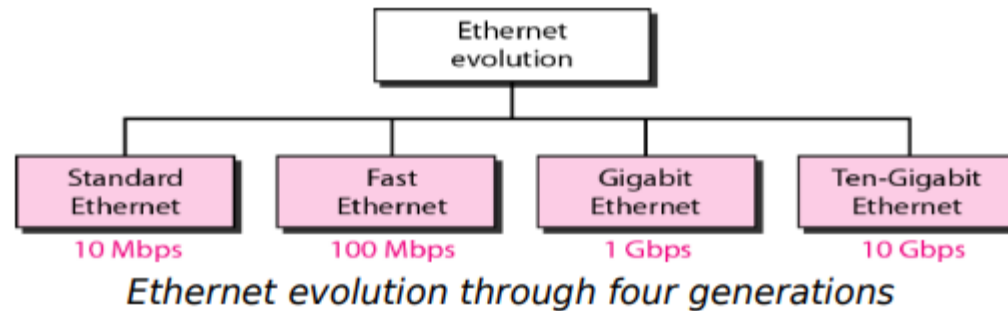
- **Wired LANs: Ethernet** In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.
- Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols. The relationship of the 802 Standard to the traditional OSI model is shown in below Figure.
- The IEEE has subdivided the data link layer into two sub layers: logical link control (LLC) and media access control). IEEE has also created several physical layer standards for different LAN protocols

- **Wired LANs: Ethernet**
- **STANDARD ETHERNET** The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through four generations. Standard Ethernet (10 Mbps), Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and Ten-Gigabit Ethernet (10 Gbps), We briefly discuss the Standard (or traditional) Ethernet in this section



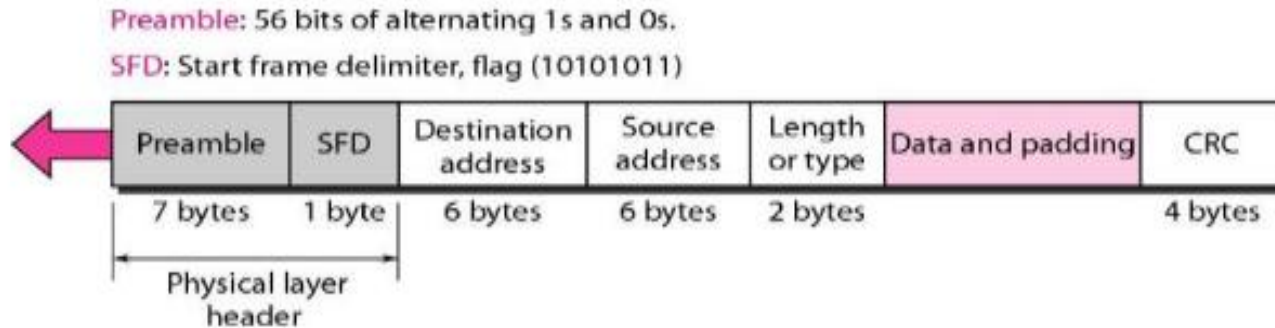
IEEE standard for LANs

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MAC Sublayer
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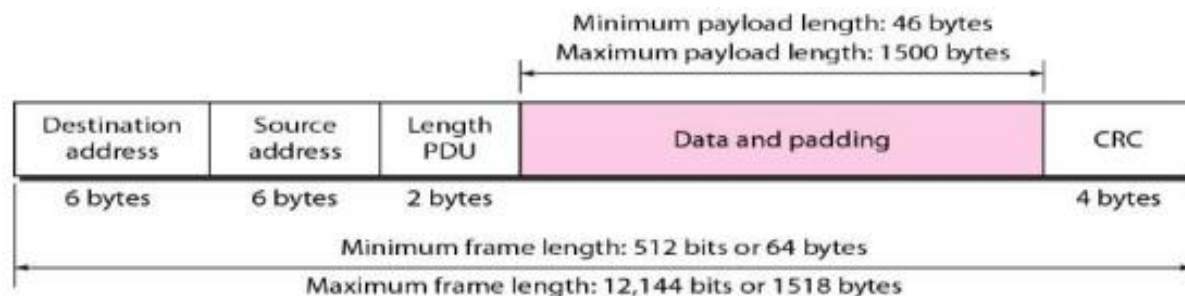
- **Frame Format** The Ethernet frame contains seven fields: preamble, SFD, DA, SA, length or type of protocol data unit (PDU), upper-layer data, and the CRC. Ethernet does not provide any mechanism for acknowledging received frames, making it what is known as an unreliable medium. Acknowledgments must be implemented at the higher layers. The format of the MAC frame is shown in below figure.



- **Preamble.** The first field of the 802.3 frame contains 7 bytes (56 bits) of alternating 0s and 1s that alerts the receiving system to the coming frame and enables it to synchronize its input timing. The pattern provides only an alert and a timing pulse. The 56-bit pattern allows the stations to miss some bits at the beginning of the frame. The preamble is actually added at the physical layer and is not (formally) part of the frame.

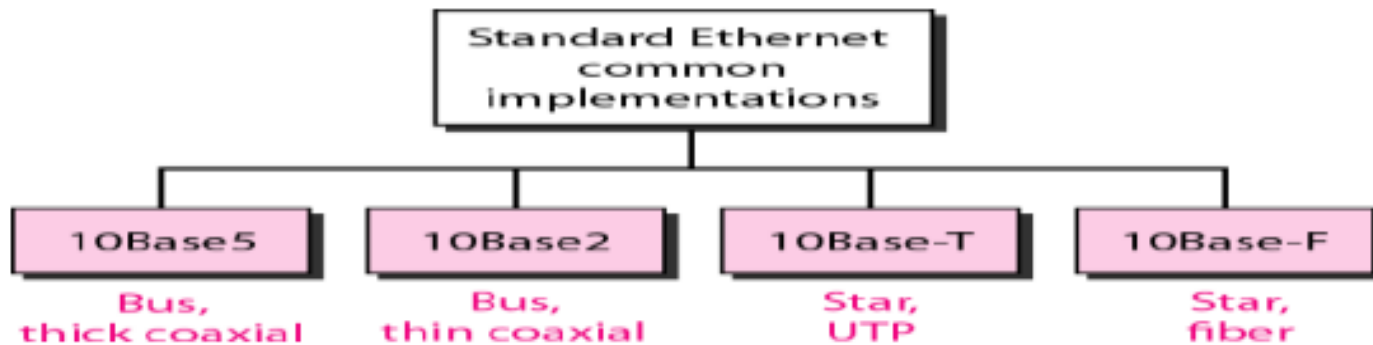
- **Start frame delimiter (SFD).** The second field (1 byte: 10101011) signals the beginning of the frame. The SFD warns the station or stations that this is the last chance for synchronization. The last 2 bits is 11 and alerts the receiver that the next field is the destination address.
- **Destination address (DA).** The DA field is 6 bytes and contains the physical address of the destination station or stations to receive the packet.
- **Source address (SA).** The SA field is also 6 bytes and contains the physical address of the sender of the packet.
- **Length or type.** This field is defined as a type field or length field. The original Ethernet used this field as the type field to define the upper-layer protocol using the MAC frame. The IEEE standard used it as the length field to define the number of bytes in the data field. Both uses are common today.
- **Data.** This field carries data encapsulated from the upper-layer protocols. It is a minimum of 46 and a maximum of 1500 bytes.
- **CRC.** The last field contains error detection information, in this case a CRC-32

- Frame Length Ethernet has imposed restrictions on both the minimum and maximum lengths of a frame, as shown in below Figure An Ethernet frame needs to have a minimum length of 512 bits or 64 bytes. Part of this length is the header and the trailer. If we count 18 bytes of header and trailer (6 bytes of source address, 6 bytes of destination address, 2 bytes of length or type, and 4 bytes of CRC), then the minimum length of data from the upper layer is $64 - 18 = 46$ bytes.
- If the upper-layer packet is less than 46 bytes, padding is added to make up the difference The standard defines the maximum length of a frame (without preamble and SFD field) as 1518 bytes. If we subtract the 18 bytes of header and trailer, the maximum length of the payload is 1500 bytes.

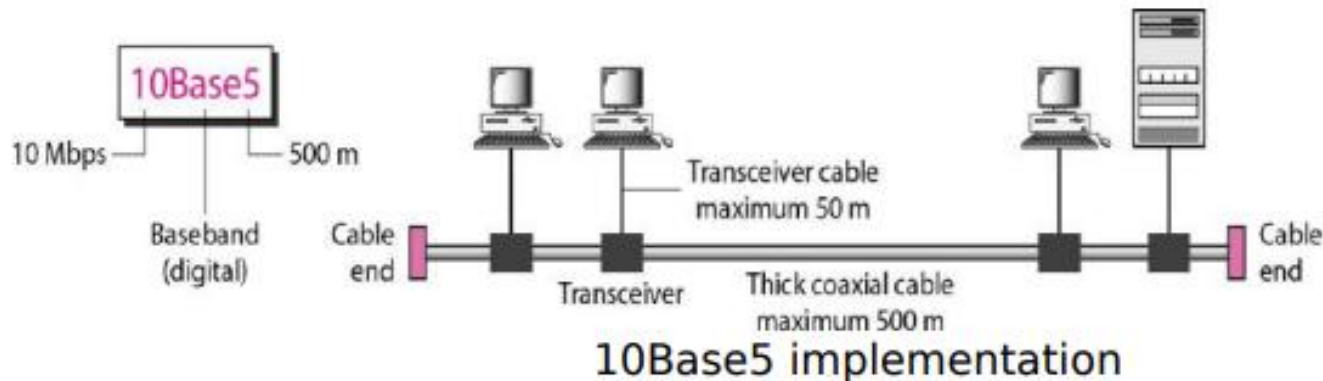


Minimum and maximum lengths

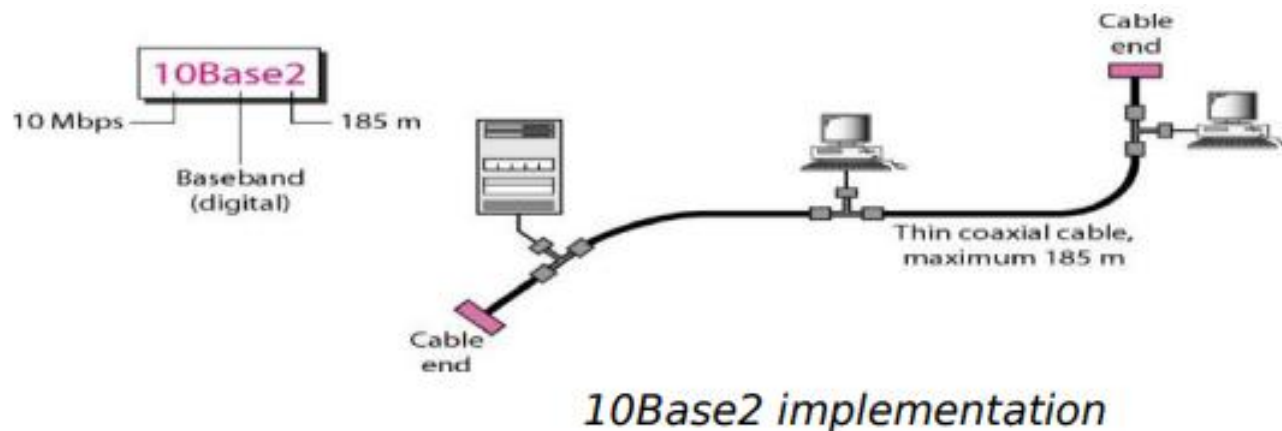
- Physical Layer The Standard Ethernet defines several physical layer implementations; four of the most common, are shown in Figure



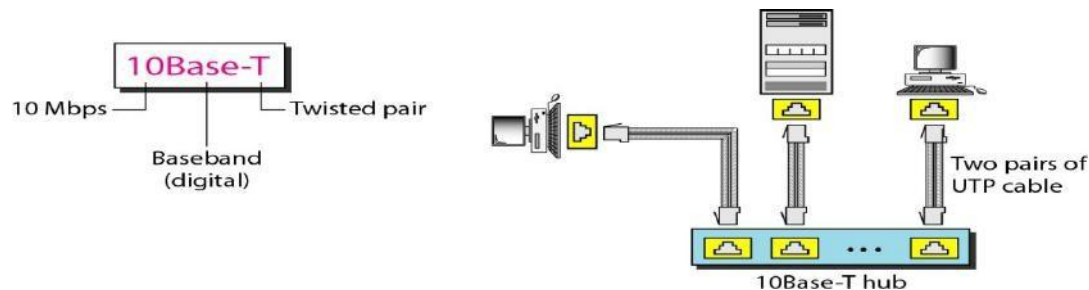
- **10Base5:** Thick Ethernet The first implementation is called 10Base5, thick Ethernet, or Thicknet. 10Base5 was the first Ethernet specification to use a bus topology with an external transceiver (transmitter/receiver) connected via a tap to a thick coaxial cable. Figure shows a schematic diagram of a 10Base5 implementation



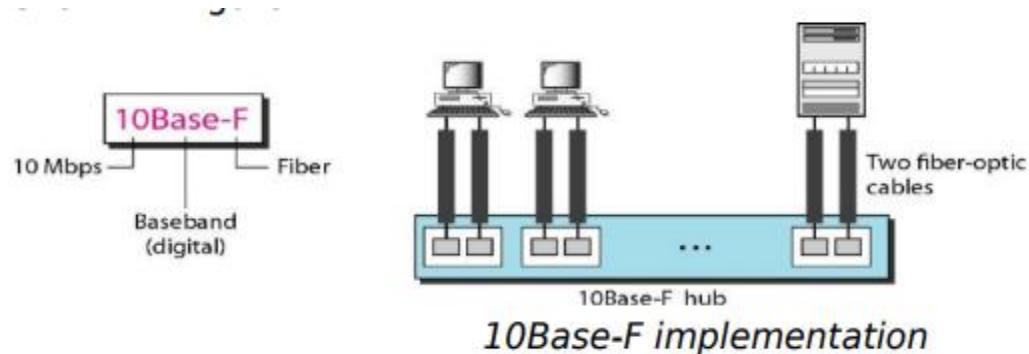
- **10Base2:** Thin Ethernet The second implementation is called 10 Base2, thin Ethernet, or Cheapernet. 10Base2 also uses a bus topology, but the cable is much thinner and more flexible. Figure shows the schematic diagram of a 10Base2 implementation. thin coaxial cable is less expensive than thick coaxial. Installation is simpler because the thin coaxial cable is very flexible. However, the length of each segment cannot exceed 185 m (close to 200 m) due to the high level of attenuation in thin coaxial cable.



- **10Base-T: Twisted-Pair Ethernet**
- The third implementation is called 10Base-T or twisted-pair Ethernet. It uses a physical star topology. The stations are connected to a hub via two pairs of twisted cable, as shown in Figure
- The maximum length of the twisted cable here is defined as 100 m, to minimize the effect of attenuation in the twisted cable
- Although there are several types of optical fiber 10-Mbps Ethernet, the most common is called 10Base-F. 10Base-F uses a star topology to connect stations to a hub. The stations are connected to the hub using two fiber-optic cables, as shown in Figure
- 10Base-F implementation

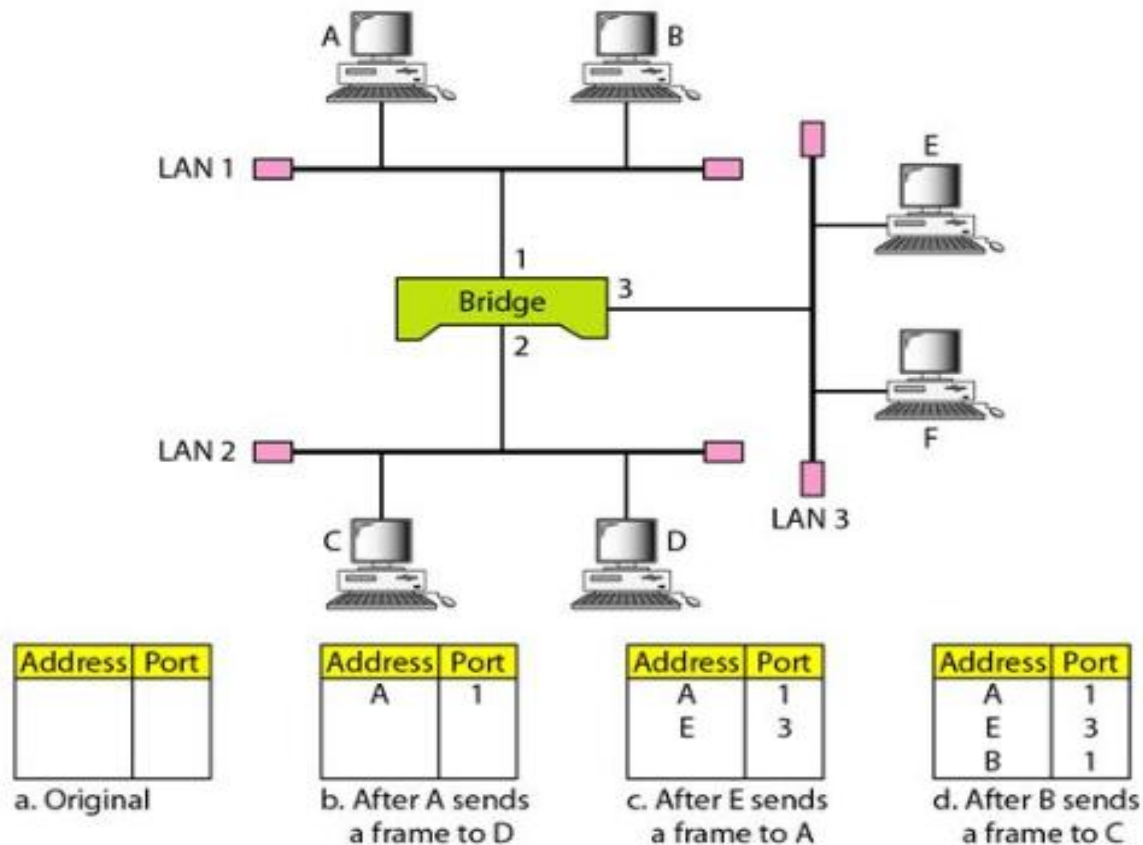


- **10Base-F implementation**
- Although there are several types of optical fiber 10-Mbps Ethernet, the most common is called 10Base-F. 10Base-F uses a star topology to connect stations to a hub. The stations are connected to the hub using two fiber-optic cables, as shown in Figure 10Base-F implementation



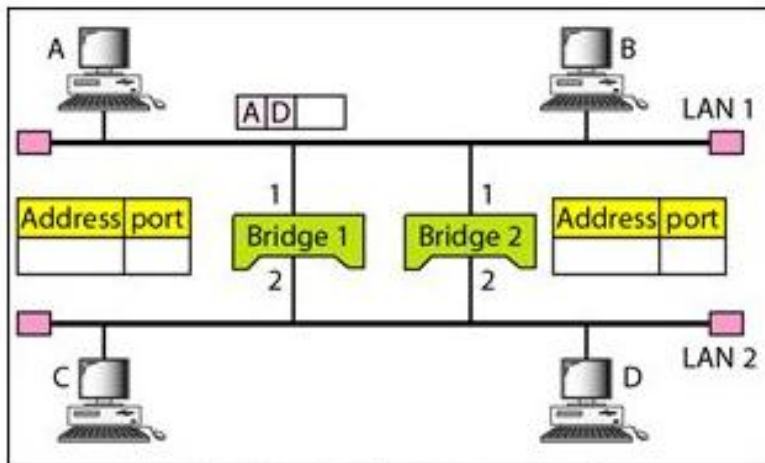
- ***Learning Bridges***
- The earliest bridges had forwarding tables that were static. The systems administrator would manually enter each table entry during bridge setup. Although the process was simple, it was not practical. If a station was added or deleted, the table had to be modified manually. The same was true if a station's MAC address changed, which is not a rare event. For example, putting in a new network card means a new MAC address.
- A better solution to the static table is a dynamic table that maps addresses to ports automatically. To make a table dynamic, we need a bridge that gradually learns from the frame movements. To do this, the bridge inspects both the destination and the source addresses.

- The destination address is used for the forwarding decision (table lookup); the source address is used for adding entries to the table and for updating purposes. Let us elaborate on this process by using the following figure

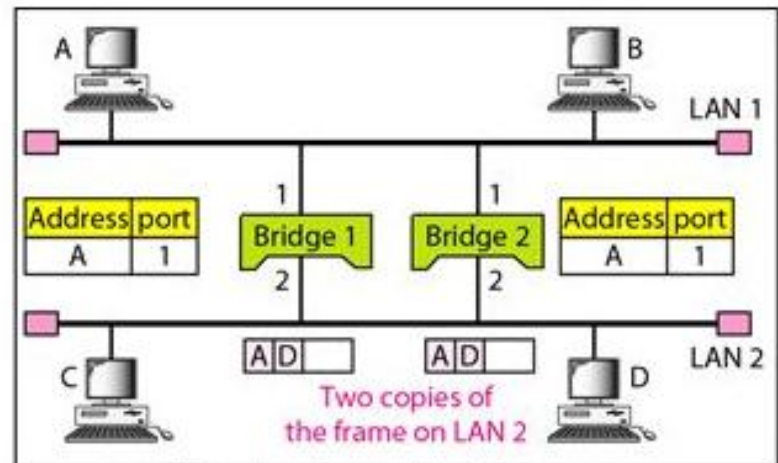


1. When station A sends a frame to station D, the bridge does not have an entry for either D or A. The frame goes out from all three ports; the frame floods the network. However, by looking at the source address, the bridge learns that station A must be located on the LAN connected to port 1. This means that frames destined for A, in the future, must be sent out through port 1. The bridge adds this entry to its table. The table has its first entry now.
2. When station E sends a frame to station A, the bridge has an entry for A, so it forwards the frame only to port 1. There is no flooding. In addition, it uses the source address of the frame, E, to add a second entry to the table.
3. When station B sends a frame to C, the bridge has no entry for C, so once again it floods the network and adds one more entry to the table.
4. The process of learning continues as the bridge forwards frames.

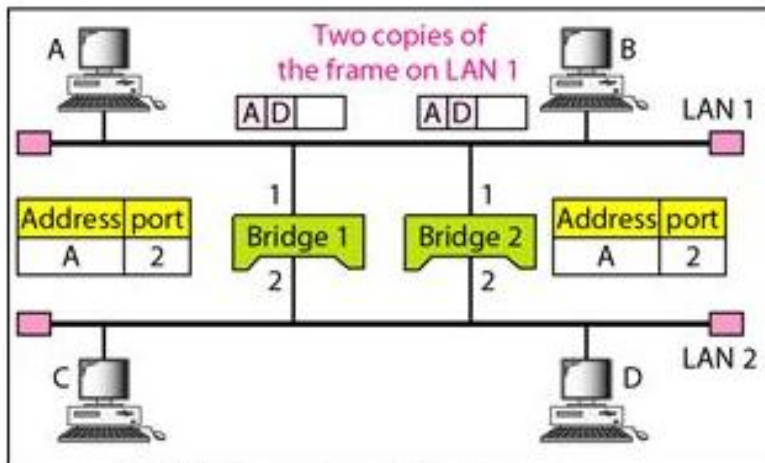
- **Loop Problem:**
- Transparent bridges work fine as long as there are no redundant bridges in the system.
- Systems administrators, however, like to have redundant bridges (more than one bridge between a pair of LANs) to make the system more reliable.
- If a bridge fails, another bridge takes over until the failed one is repaired or replaced. Redundancy can create loops in the system, which is very undesirable.
- The following figure shows a very simple example of a loop created in a system with two LANs connected by two bridges.



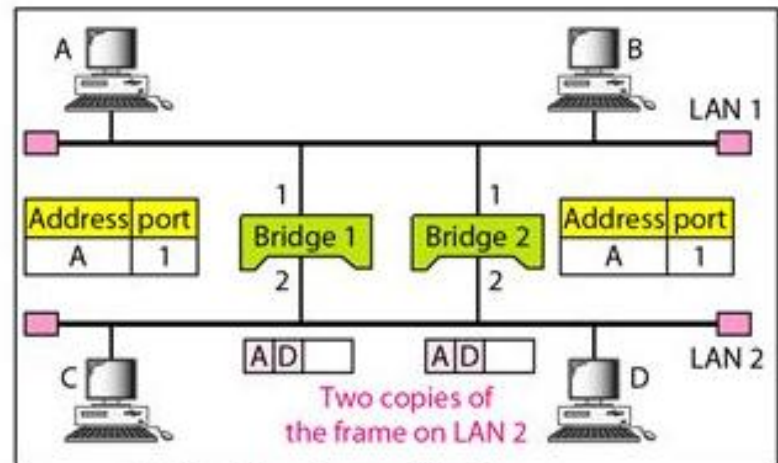
a. Station A sends a frame to station D



b. Both bridges forward the frame



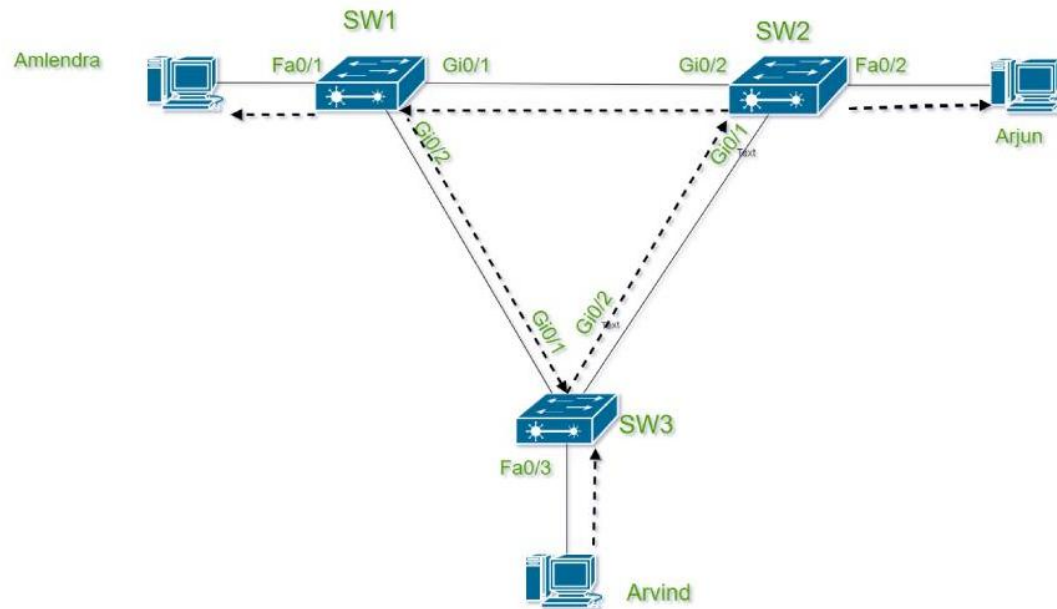
c. Both bridges forward the frame



d. Both bridges forward the frame

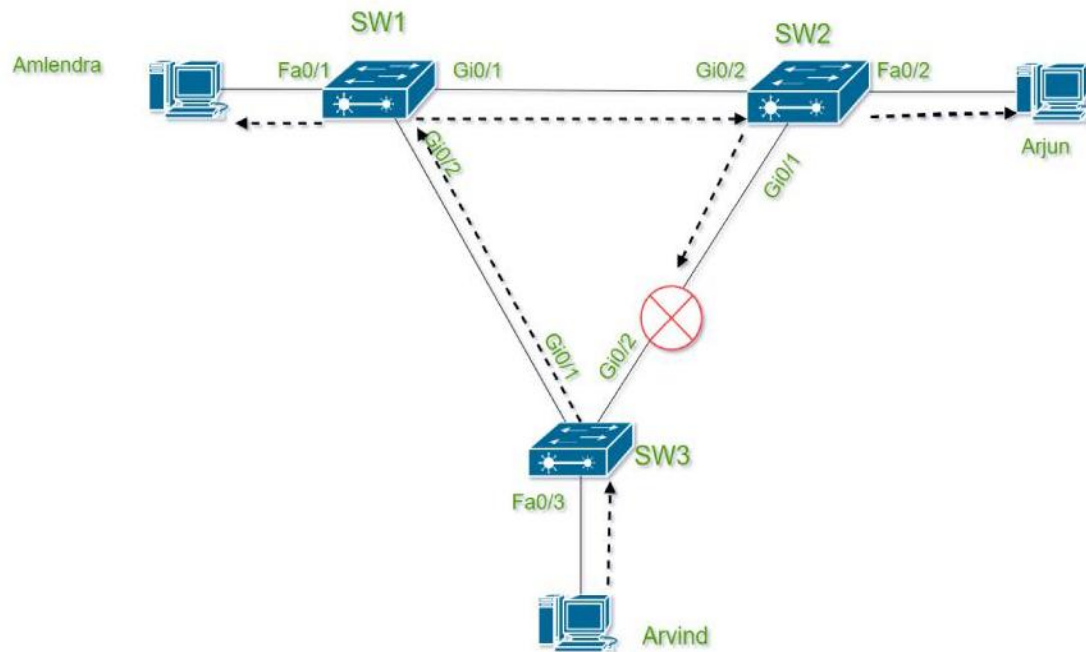
1. Station A sends a frame to station D. The tables of both bridges are empty. Both forward the frame and update their tables based on the source address A.
2. Now there are two copies of the frame on LAN 2. The copy sent out by bridge 1 is received by bridge 2, which does not have any information about the destination address D; it floods the bridge. The copy sent out by bridge 2 is received by bridge 1 and is sent out for lack of information about D. Note that each frame is handled separately because bridges, as two nodes on a network sharing the medium, use an access method such as CSMA/CD. The tables of both bridges are updated, but still there is no information for destination D.
3. Now there are two copies of the frame on LAN 1. Step 2 is repeated, and both copies flood the network.
4. The process continues on and on. Note that bridges are also repeaters and regenerate frames. So in each iteration, there are newly generated fresh copies of the frames.

- **Spanning Tree Protocol**
- In a LAN, redundant links are added to improve the network availability of LAN. But these redundant links may cause the frame to loop in the network for an infinite time until some action is taken, e.g, some links are taken down. To cope with the problem of frame looping, Spanning Tree Protocol (STP) comes into play.
- **Need for Spanning Tree:**
- Consider the scenario below with 3 switches with one user attached to each switch.



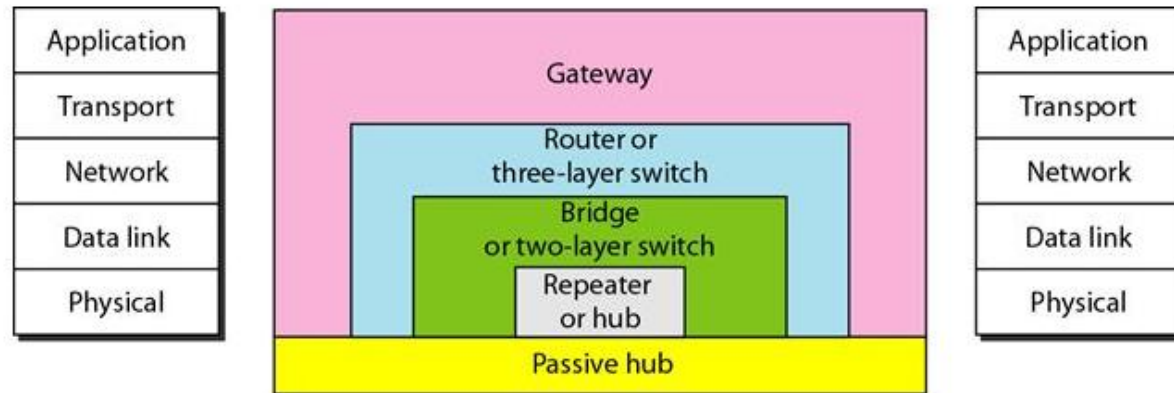
- Arvind sends a broadcast frame to LAN and as nature of switch frame are send out from other ports (Gi0/1 & Gi0/2) except receiving port (Fa0/3). Now, this frame goes to SW2, SW2 also broadcast frame out of Gi0/2 and Fa0/2 ports.
- SW1 receives frame in its Gi0/1 ports. SW1 also broadcast frame then this frame goes to SW3 and frame broadcasting goes on.
- Remember that this frame broadcast also occurs in other direction from SW3 out of Gi0/1 port. Above discussed frame looping was from SW3's Gi0/2 ports.
- You can imagine the frame flooding in that small LAN. This forever looping of frames around LAN is called *Broadcast storm*. This Looping of frames causes three problems as stated below:
- **MAC table instability** - Due to looping of frame around LAN, MAC-Table of switch get changed frequently. Looping causes incorrect MAC-table entries resulting in incorrect frame delivery.
- **Broadcast Storm** - Repeated forwarding of frames around links in LAN causes the inefficient use of links.
- **Multiple Frame Transmission** - A very serious negative effect of looping is that multiple copies of same frame are delivered to host. This process left host with confusion.

- **How Spanning Tree Protocol Helps?**
- Spanning tree protocol prevent looping of frames around LAN by placing ports of switch in either *forwarding* or *blocking* state. Interfaces (ports of switch) which are in forwarding state act as normally but Interfaces in blocking state doesn't process any frame received except STP messages and other important overheads. Blocking Interfaces doesn't learn MAC addresses, don't forward frames and don't process received frames. Now if we again consider the above discussed scenario with SW3's Gi0/2 interface in blocking state.



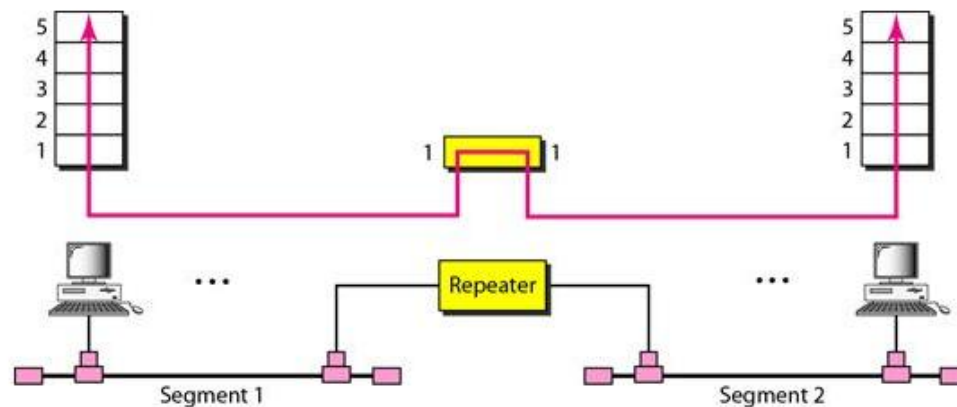
- Arvind sends the frame to SW3.
- SW3 forwards frame only to Gi0/1 port as Gi0/2 port is in blocking state.
- Now SW1 receives frame and forward to Fa0/1 and Gi0/1 interfaces.
- SW2 receives frame and forward to Fa0/2 and Gi0/1 interfaces.
- SW3 will receive the frame on Gi0/2 interface but ignores the frame as it is in blocking state.

- **Connecting Devices** The five categories contain devices which can be defined as:

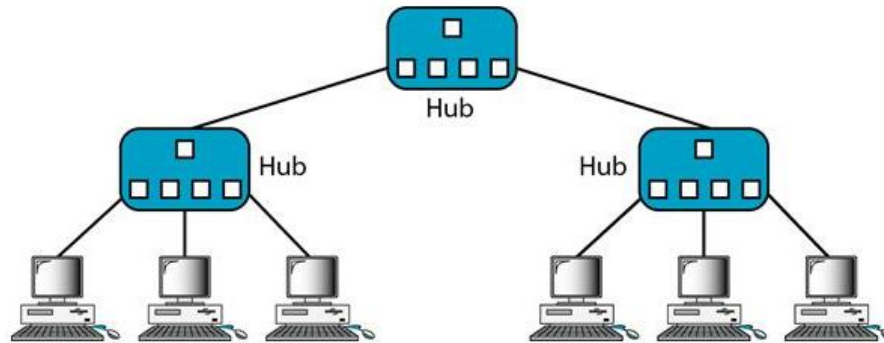


- Those which operate below the physical layer such as a passive hub.
- Those which operate at the physical layer (a repeater or an active hub).
- Those which operate at the physical and data link layers (a bridge or a two-layer switch).
- Those which operate at the physical, data link, and network layers (a router or a three-layer switch).
- Those which can operate at all five layers (a gateway).

- ***Passive Hubs:***
- A passive hub is just a connector. It connects the wires coming from different branches. In a star-topology Ethernet LAN, a passive hub is just a point where the signals coming from different stations collide; the hub is the collision point. Its location in the Internet model is below the physical layer.
- ***Repeaters***
- A repeater is a device that operates only in the physical layer. Signals that carry information within a network can travel a fixed distance before attenuation endangers the integrity of the data. A repeater receives a signal and, before it becomes too weak or corrupted, regenerates the original bit pattern. The repeater then sends the refreshed signal. A repeater can extend the physical length of a LAN, as shown in the following figure.

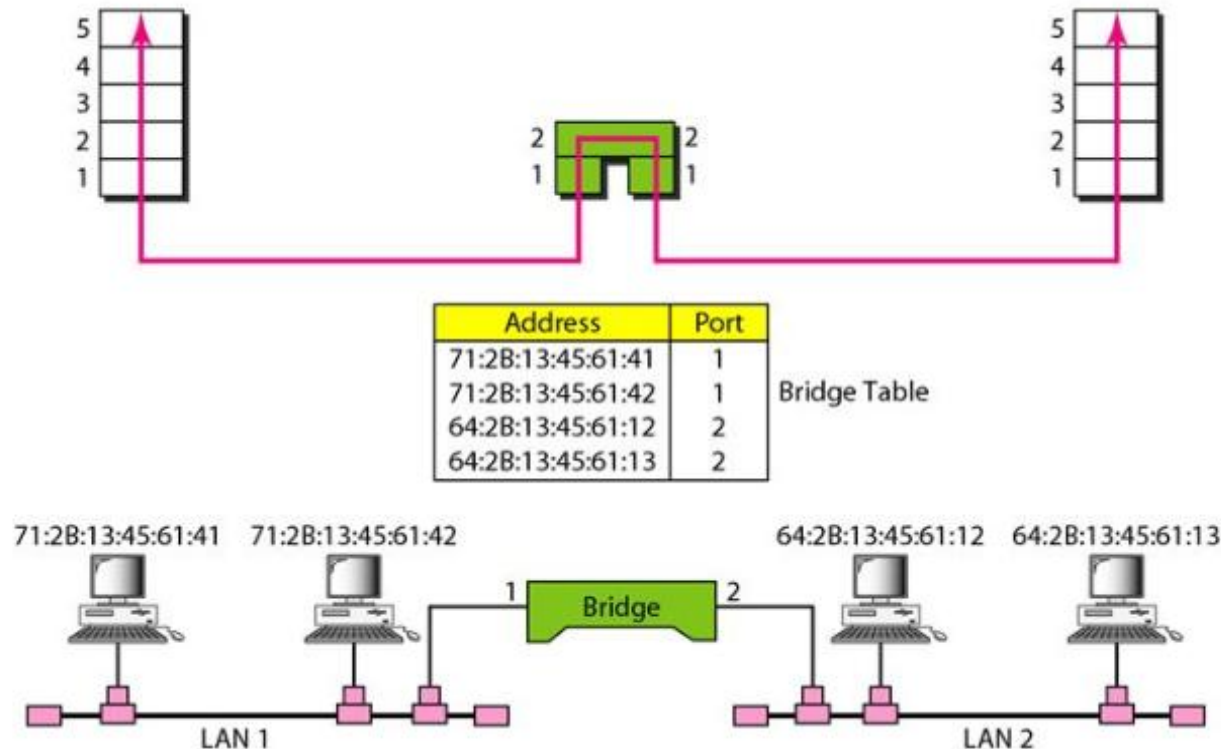


- **Active Hubs:**
- An active hub is actually a multipart repeater. It is normally used to create connections between stations in a physical star topology. However, hubs can also be used to create multiple levels of hierarchy, as shown in Figure



- The difference in the functionality of a bridge and a repeater is, a bridge has filtering capability. It can check the destination address of a frame and decide if the frame should be forwarded or dropped. If the frame is to be forwarded, the decision must specify the port. A bridge has a table that maps addresses to ports.

- **Bridge** :a bridge has filtering capability. It can check the destination address of a frame and decide if the frame should be forwarded or dropped. If the frame is to be forwarded, the decision must specify the port. A bridge has a table that maps addresses to ports.



- In the above figure, two LANs are connected by a bridge. If a frame destined for station 712B13456142 arrives at port 1, the bridge consults its table to find the departing port. According to its table, frames for 712B13456142 leave through port 1; therefore, there is no need for forwarding, and the frame is dropped. On the other hand, if a frame for 712B13456141 arrives at port 2, the departing port is port 1 and the frame is forwarded.
- In the first case, LAN 2 remains free of traffic; in the second case, both LANs have traffic. In our example, we show a two-port bridge; in reality a bridge usually has more ports. Note also that a bridge does not change the physical addresses contained in the frame.

- ***Routers***
- A router is a three-layer device that routes packets based on their logical addresses (host-to-host addressing). A router normally connects LANs and WANs in the Internet and has a routing table that is used for making decisions about the route. The routing tables are normally dynamic and are updated using routing protocols.

